FlowchartFun

Guided

OCR GCSE Computer Science (J277) flowchart practice.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.

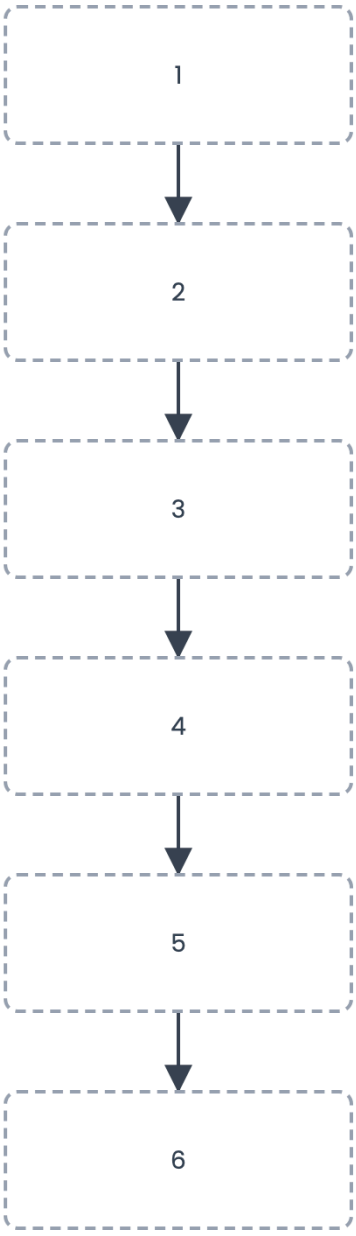
Name: _____ Class: _____

Level 1: Flowchart Basics

Easy · Guided

Create a simple flowchart that calculates the sum of two numbers. Drag each symbol into the correct place.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



End

Input A

Input B

Output Sum

Calculate Sum = A + B

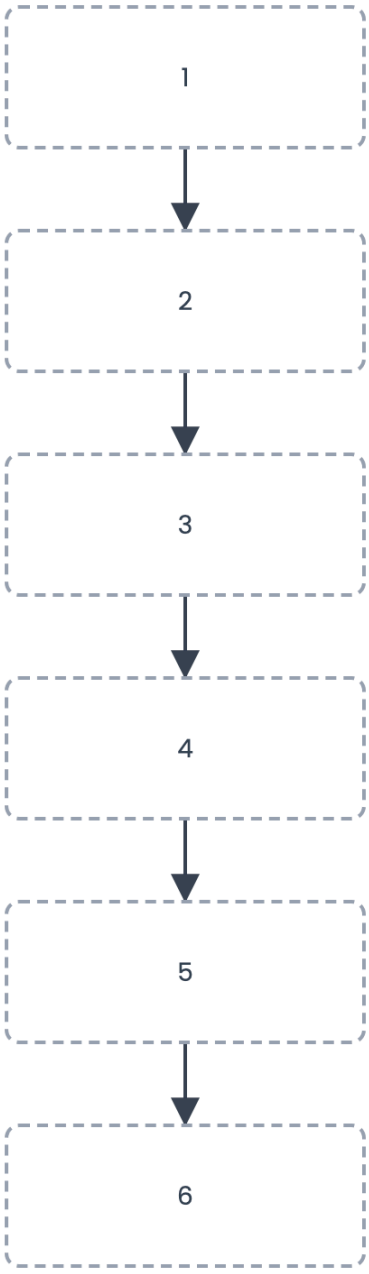
Start

Level 2: Input, Process, Output

Easy · Guided

Make a flowchart that finds the area of a rectangle. Watch out: one symbol is not needed!

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



End

Input length

Input width

Output area

area = length × width

Start

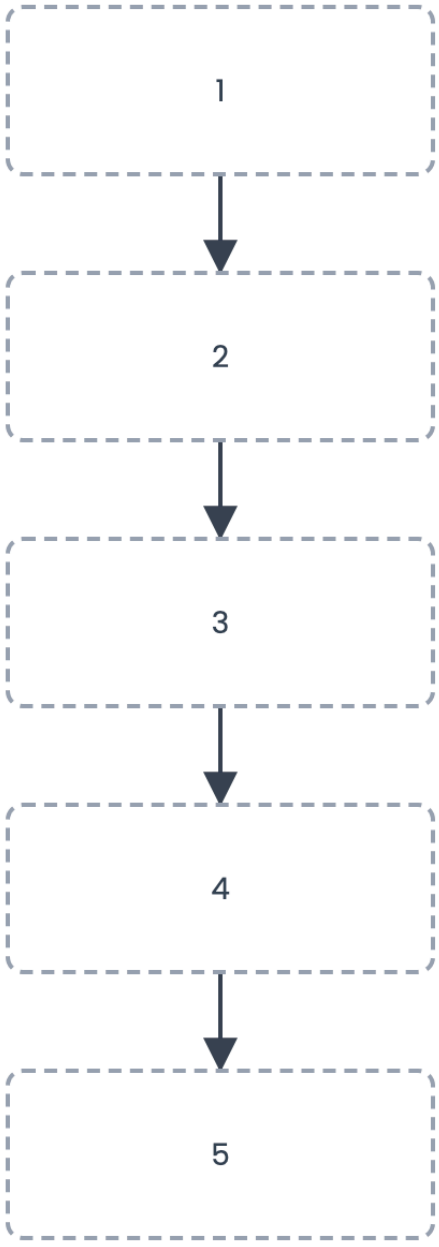
perimeter = 2 × (length + width)

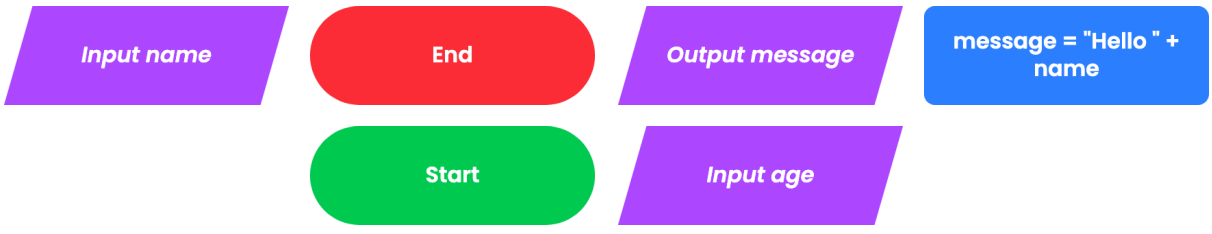
Level 3: Hello, Name

Easy · Guided

Ask for a name and greet the user. Not every symbol is needed.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



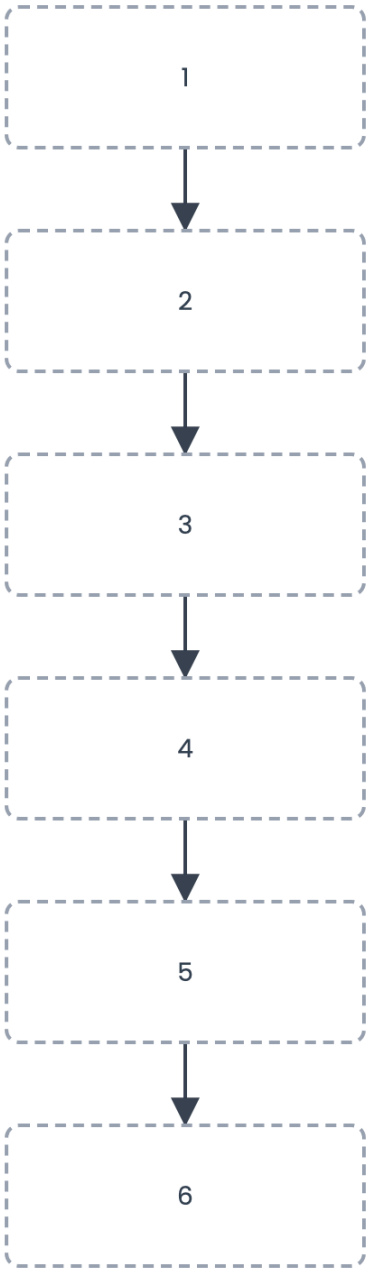


Level 4: Average of Two

Easy · Guided

Input two numbers and output their average. Remember the brackets!

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



End

Input number 1

Input number 2

Output average

average = (number 1 + number 2) / 2

Start

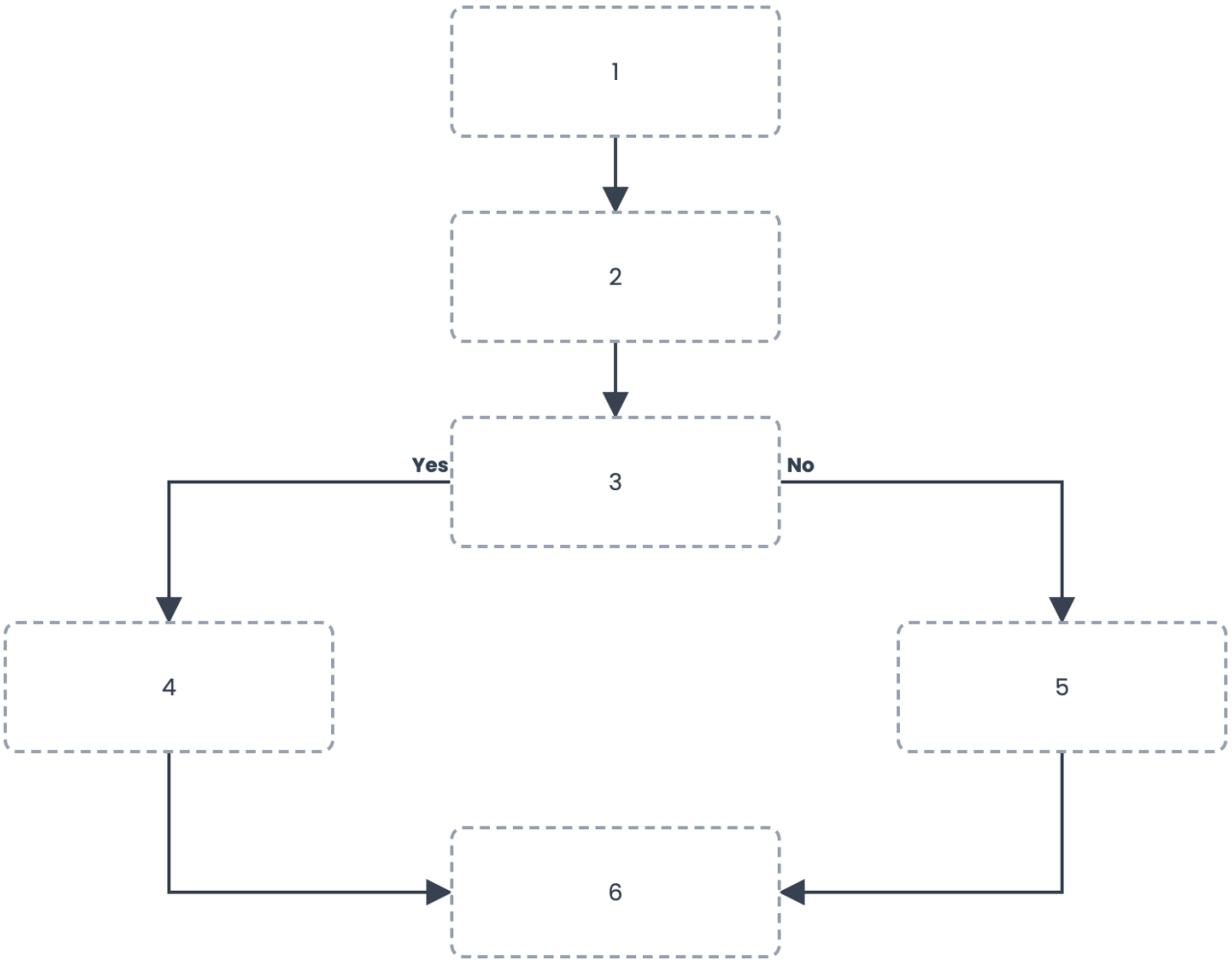
average = number 1 + number 2 / 2

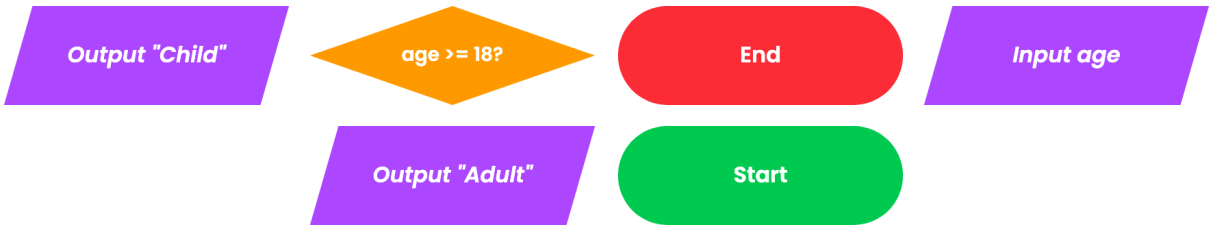
Level 5: Making a Decision

Easy · Guided

Output whether a person is an adult or a child. Remember: diamonds are decisions, with a Yes path and a No path.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



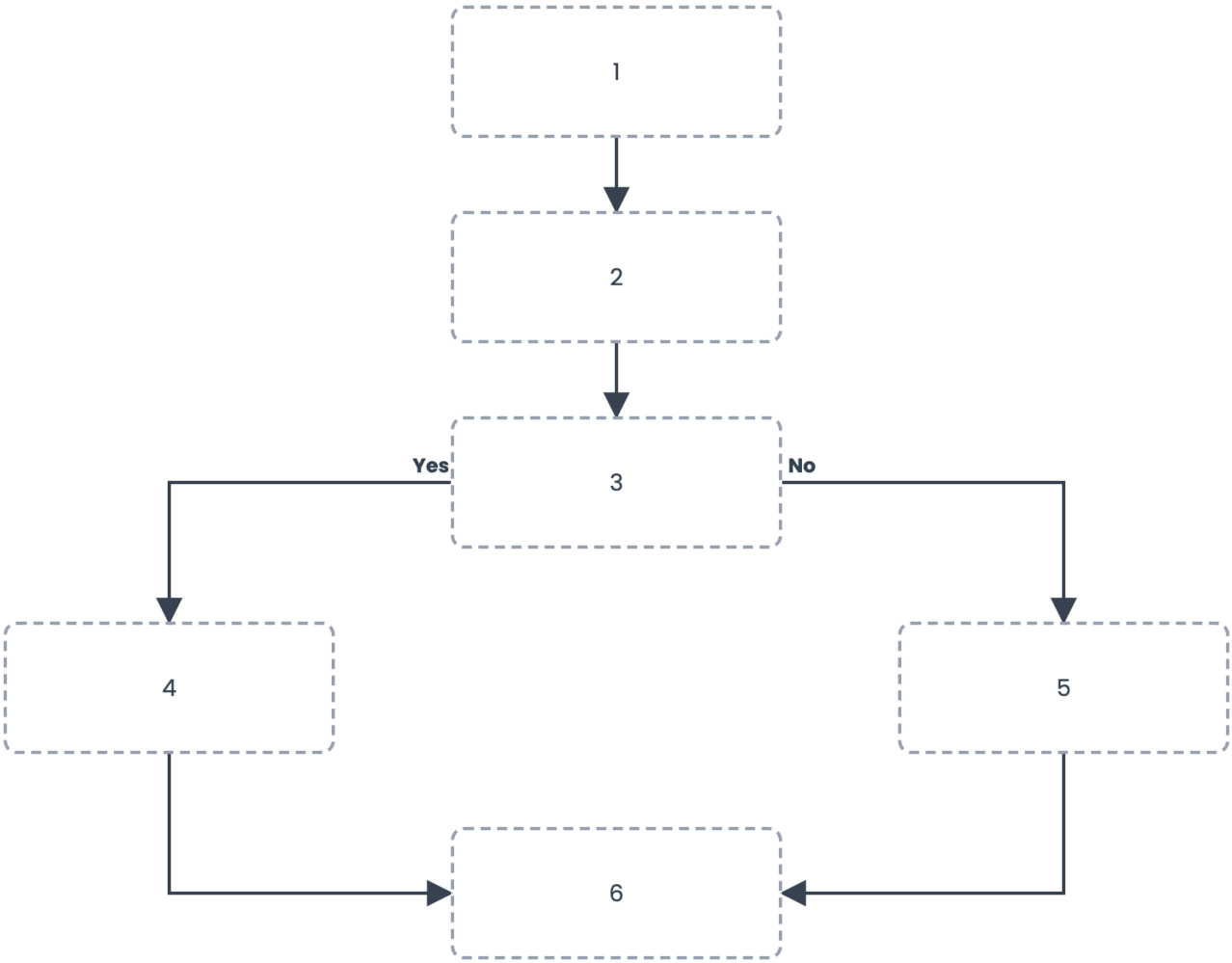


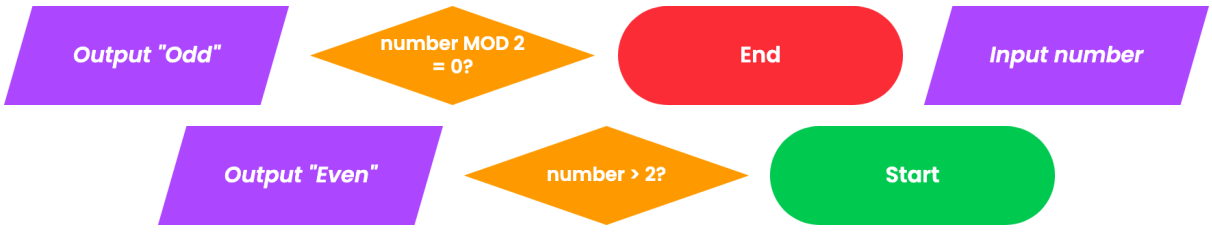
Level 6: Odd or Even

Easy · Guided

Use MOD (the remainder after division) to decide whether a number is even or odd.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



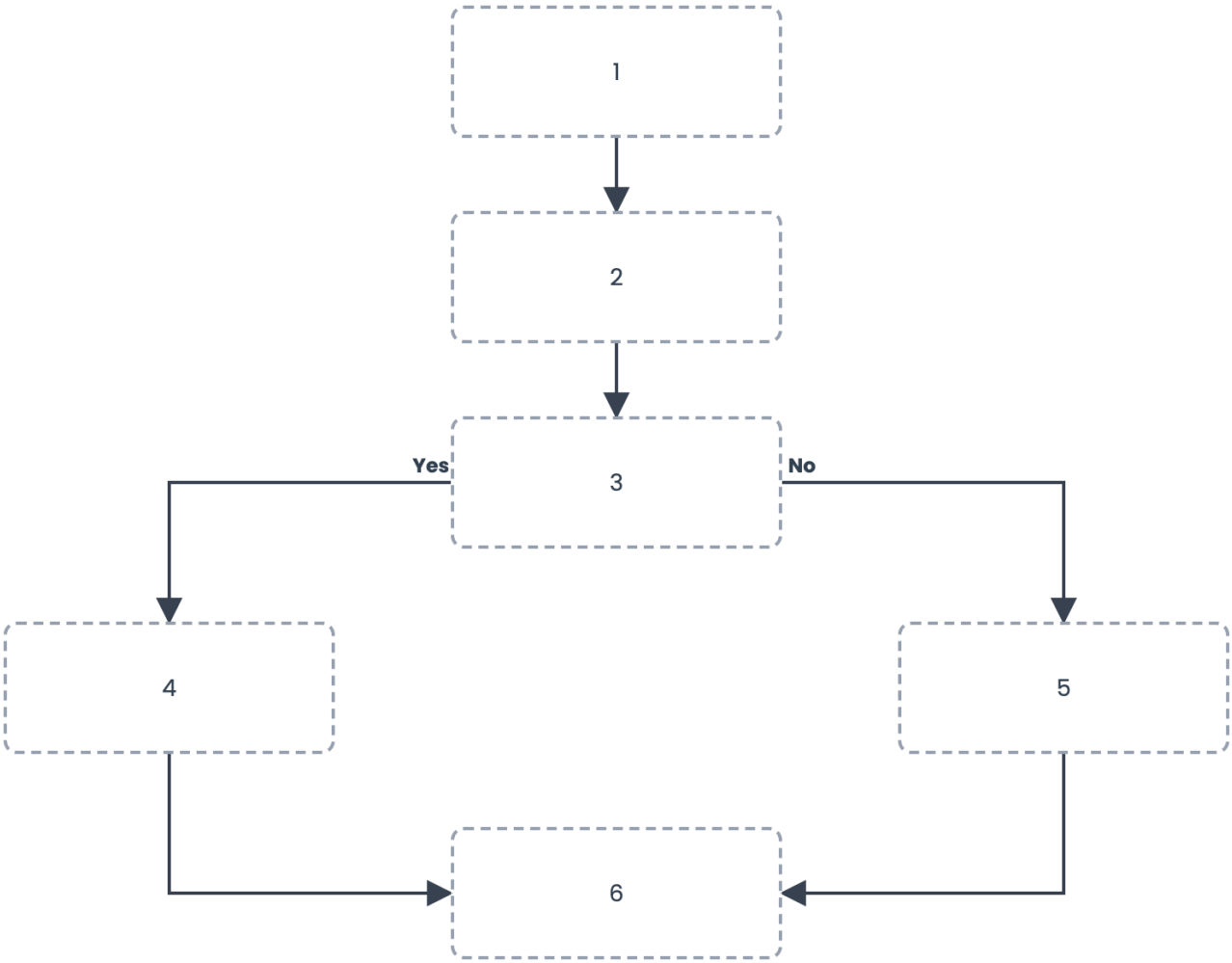


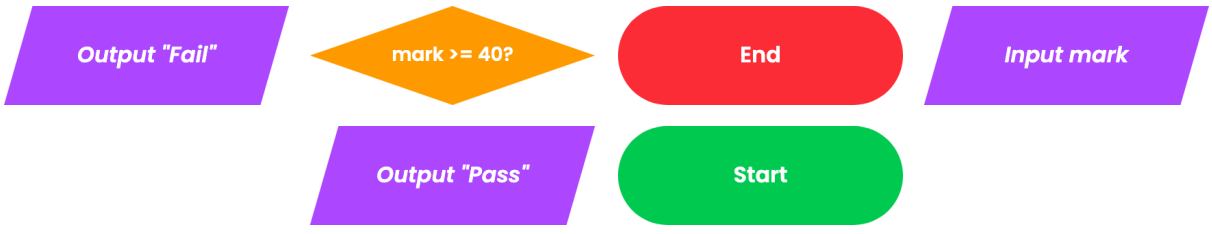
Level 7: Pass or Fail

Easy · Guided

A mark of 40 or more is a pass. Anything else is a fail.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



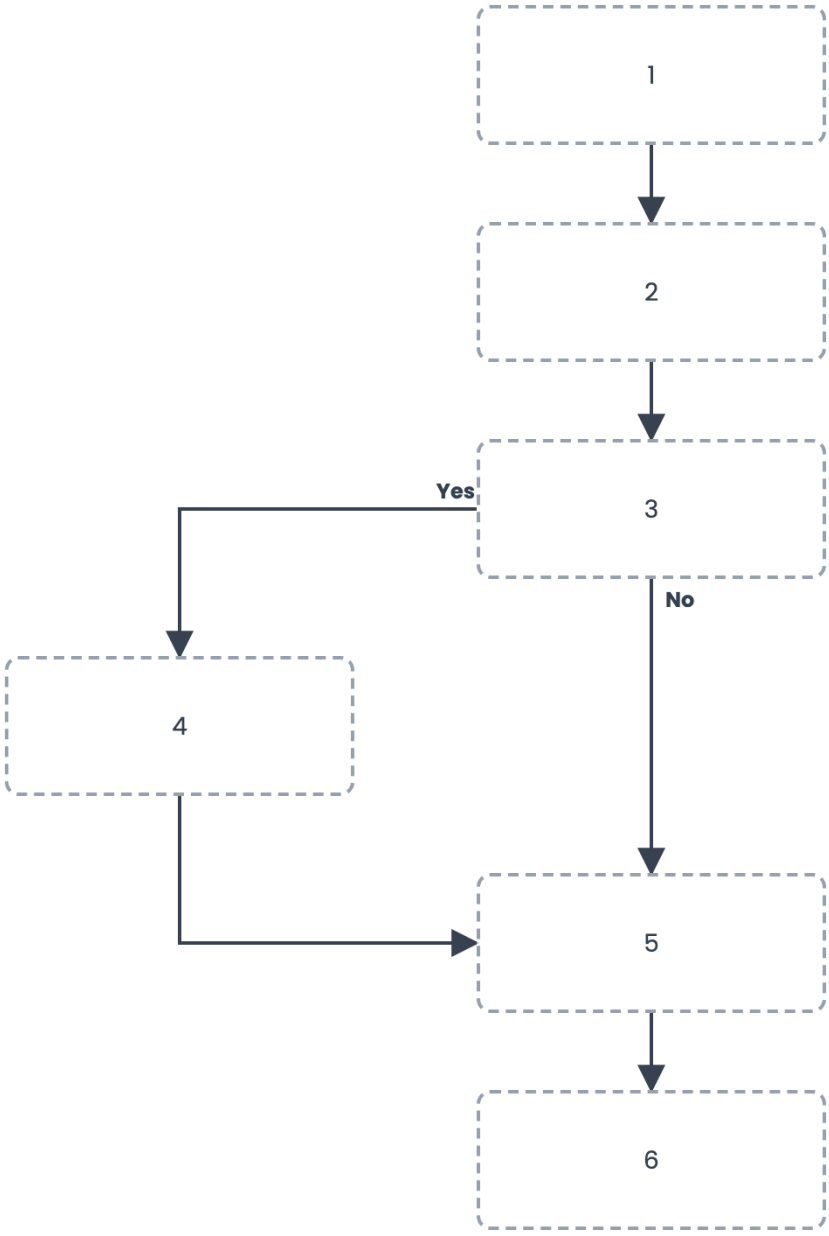


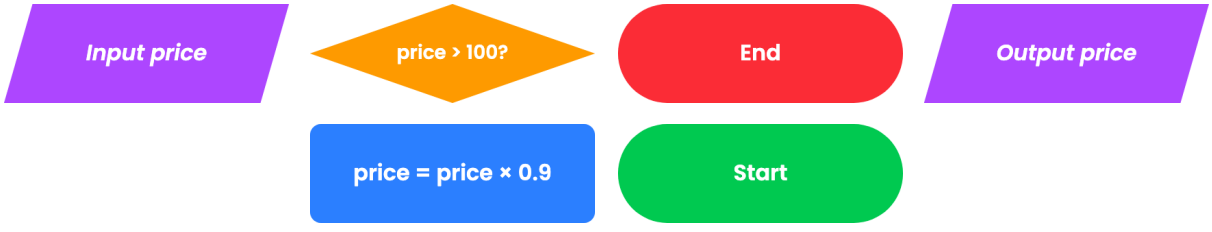
Level 8: Discount

Easy · Guided

Shoppers get 10% off if they spend more than £100. Everyone sees a price at the end. Notice that one path skips a step.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



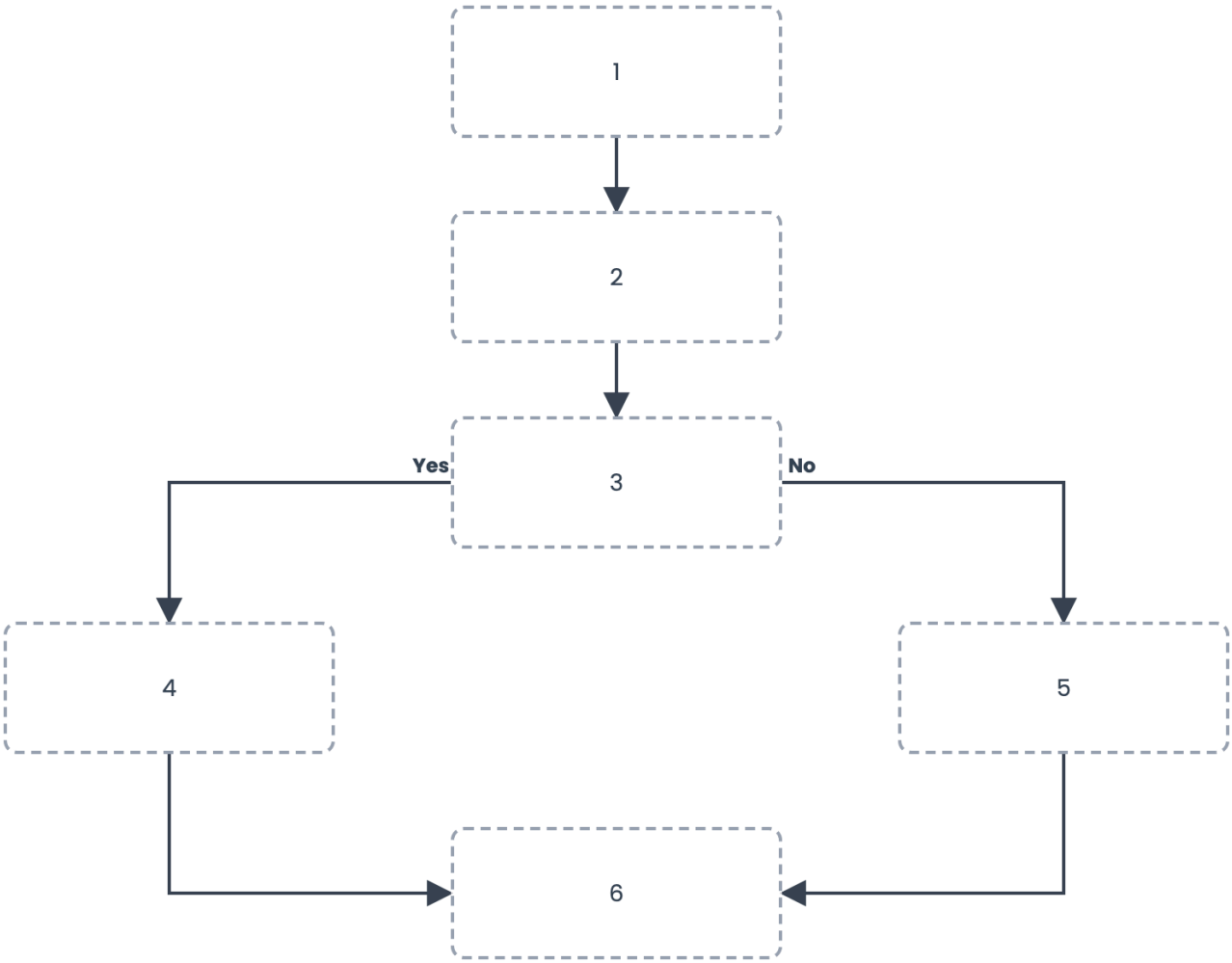


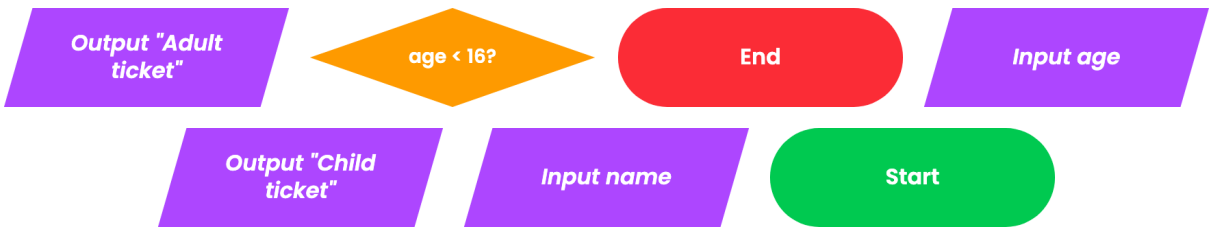
Level 9: Cinema Ticket

Easy · Guided

Under 16s pay the child price. Everyone else pays the adult price.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



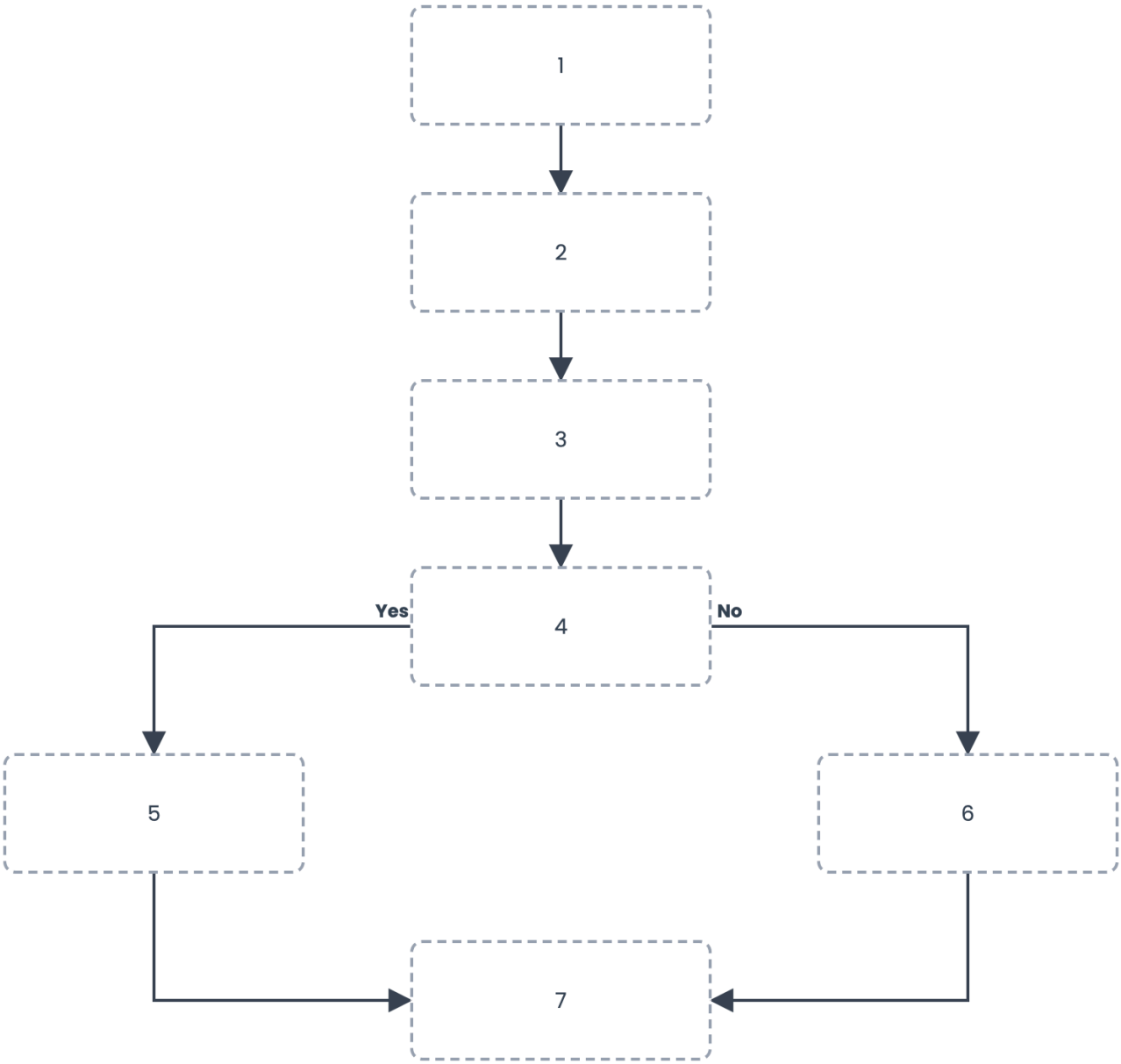


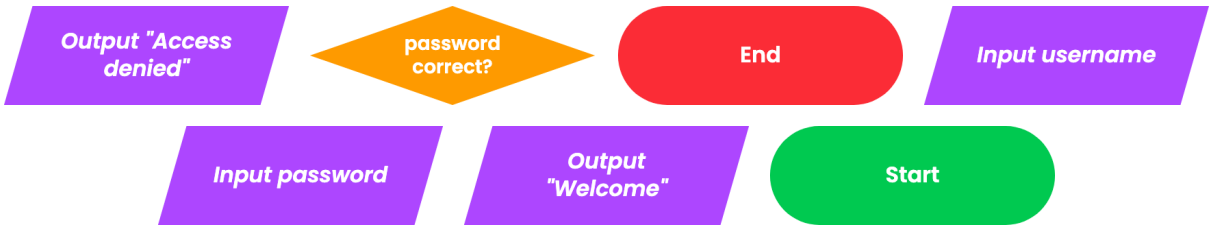
Level 10: Login

Easy · Guided

Ask for a username and password, then decide whether to let the user in.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



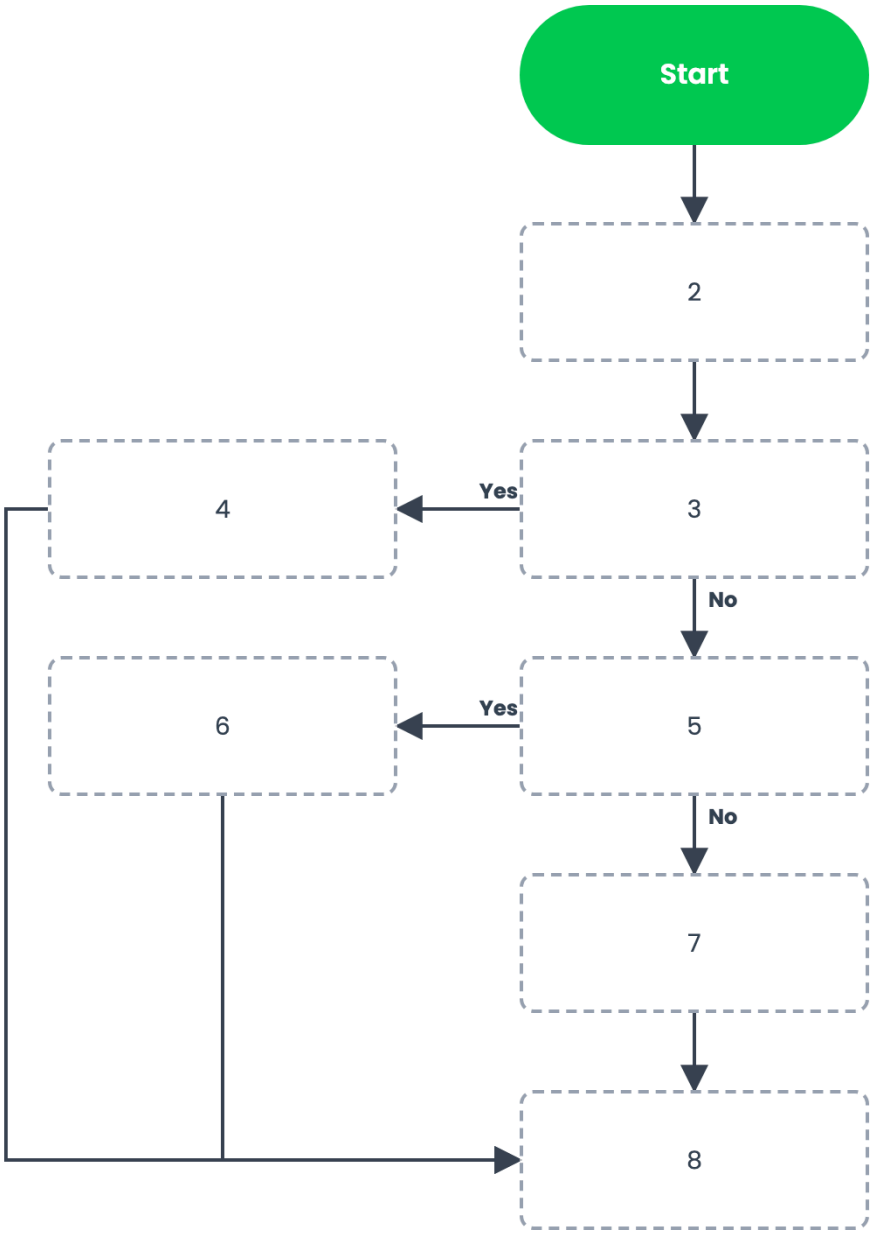


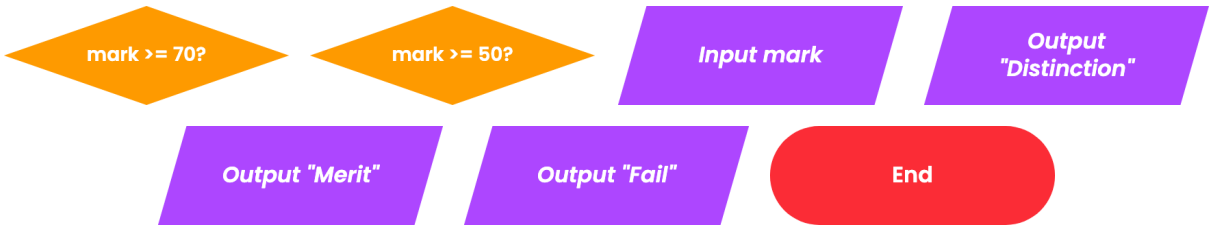
Level 11: Grade Boundaries

Medium · Guided

Output Distinction for 70 or more, Merit for 50 or more, otherwise Fail. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



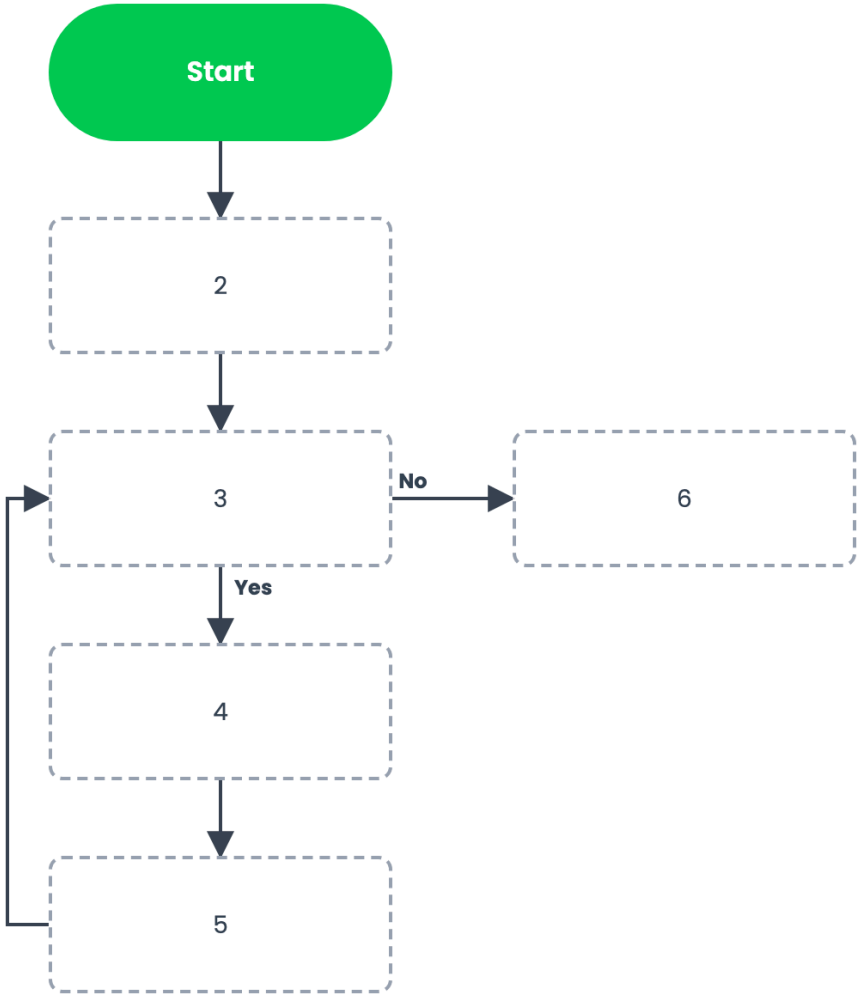


Level 12: Count to Five

Medium · Guided

Output the numbers 1 to 5. The arrow that goes back up is a loop. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



count <= 5?

End

count = count + 1

Output count

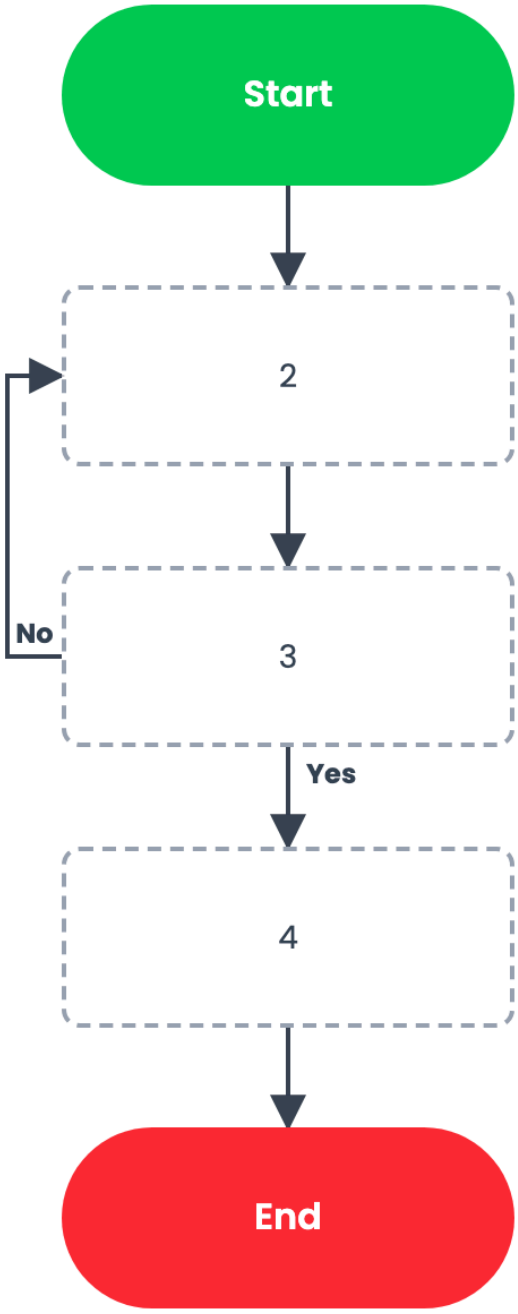
count = 1

Level 13: Password Check

Medium · Guided

Keep asking for the password until it is correct, then grant access. Start and End are placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



Input password

password correct?

Output "Access granted"

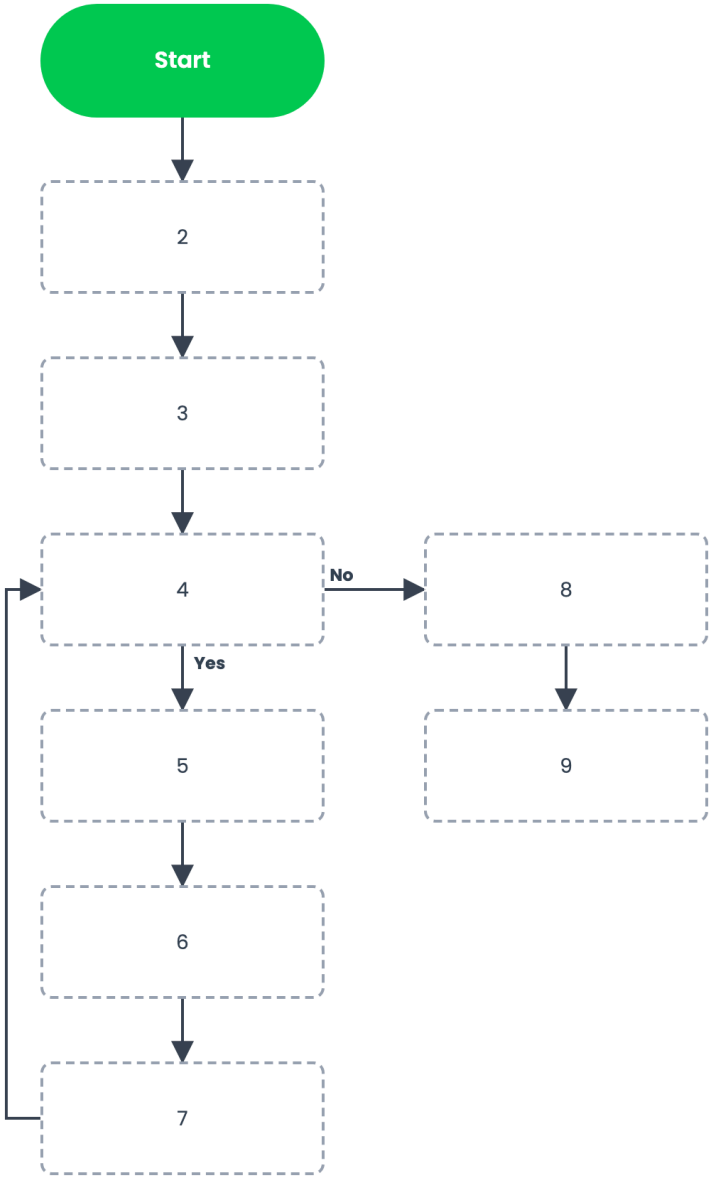
password = 0

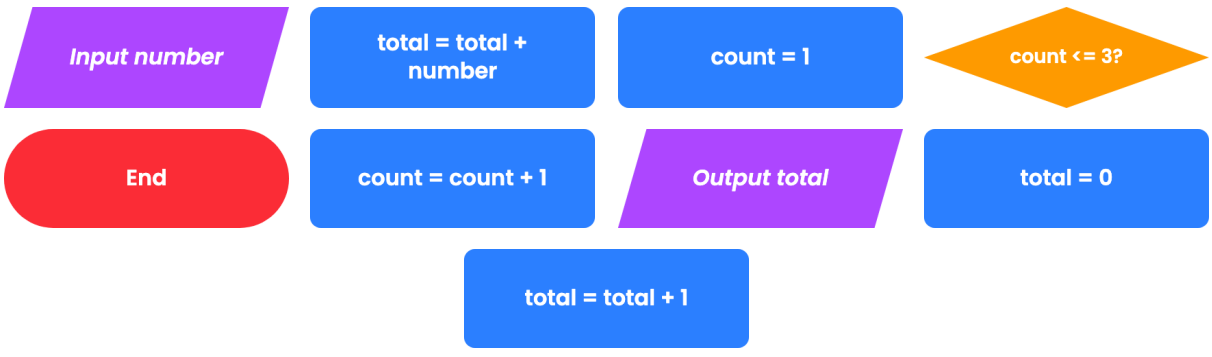
Level 14: Running Total

Medium · Guided

Input three numbers and output their total. This one combines a counter, a total and a loop.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



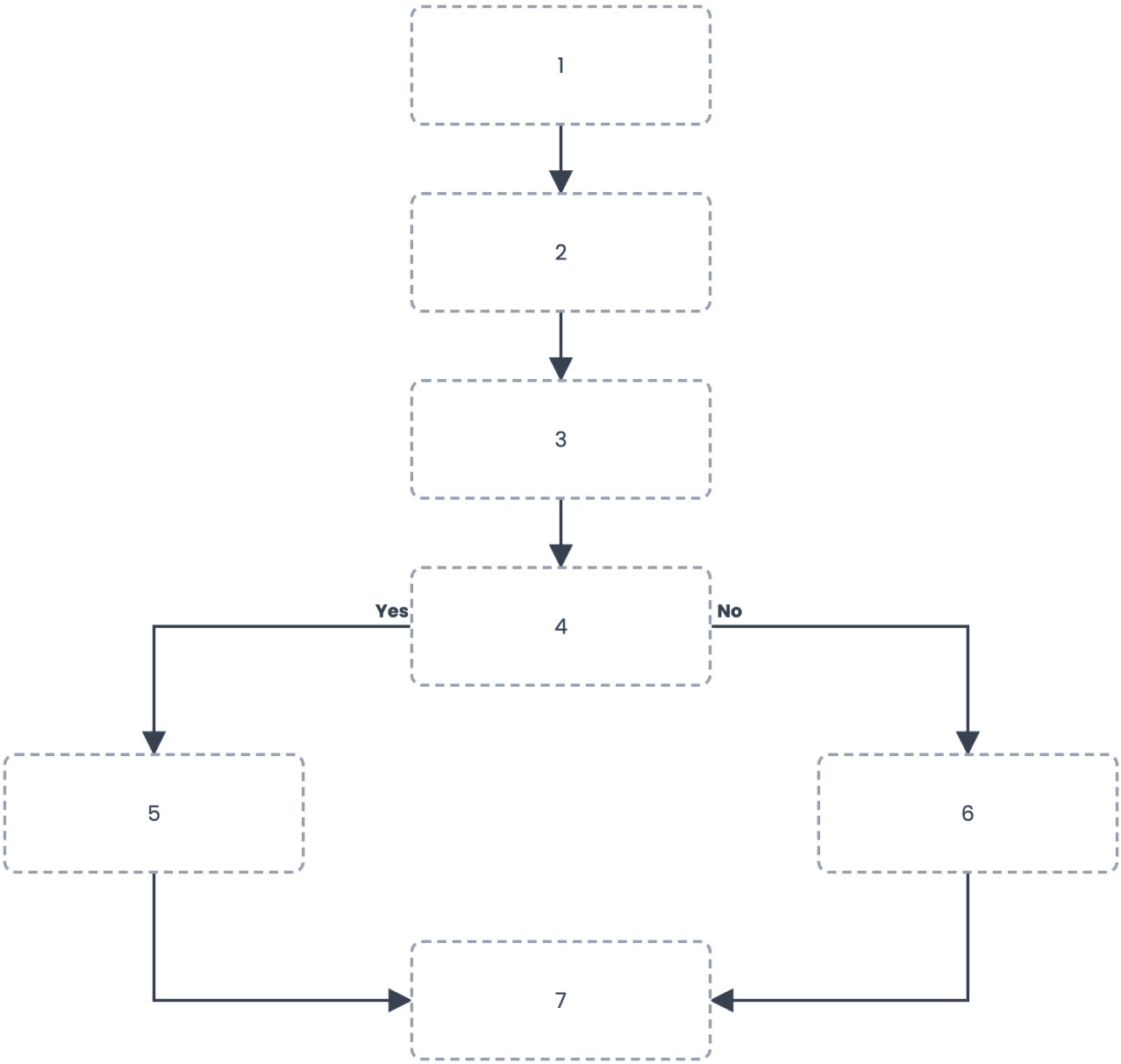


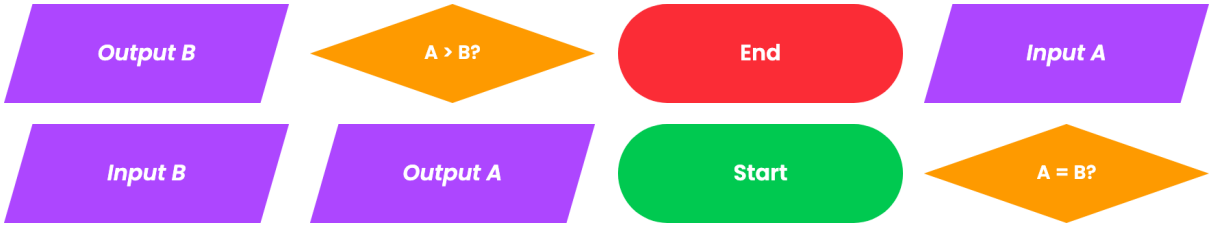
Level 15: Larger Number

Medium · Guided

Input two numbers and output the larger one.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



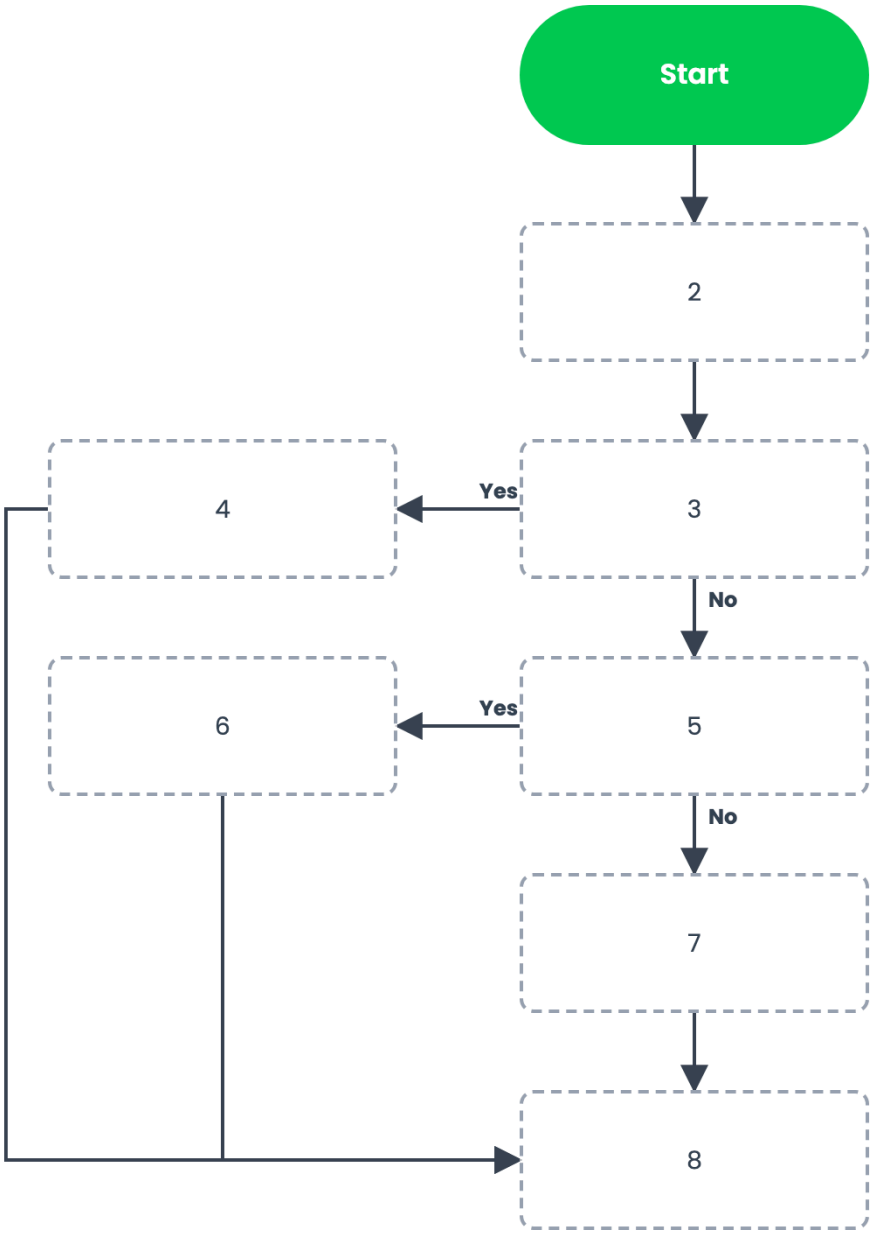


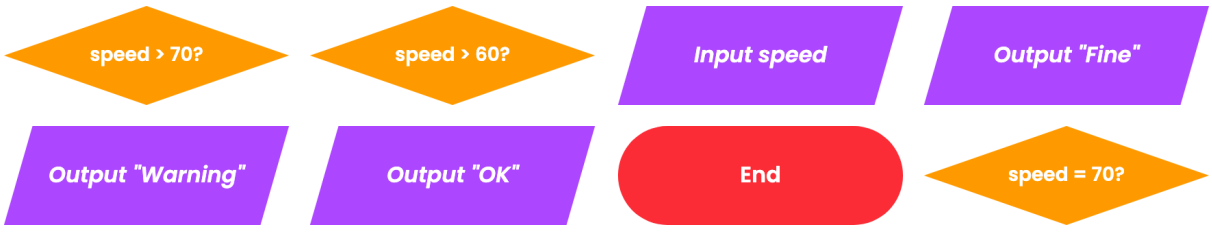
Level 16: Speed Camera

Medium · Guided

Over 70 is a fine. Over 60 is a warning. Anything else is fine to drive on. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



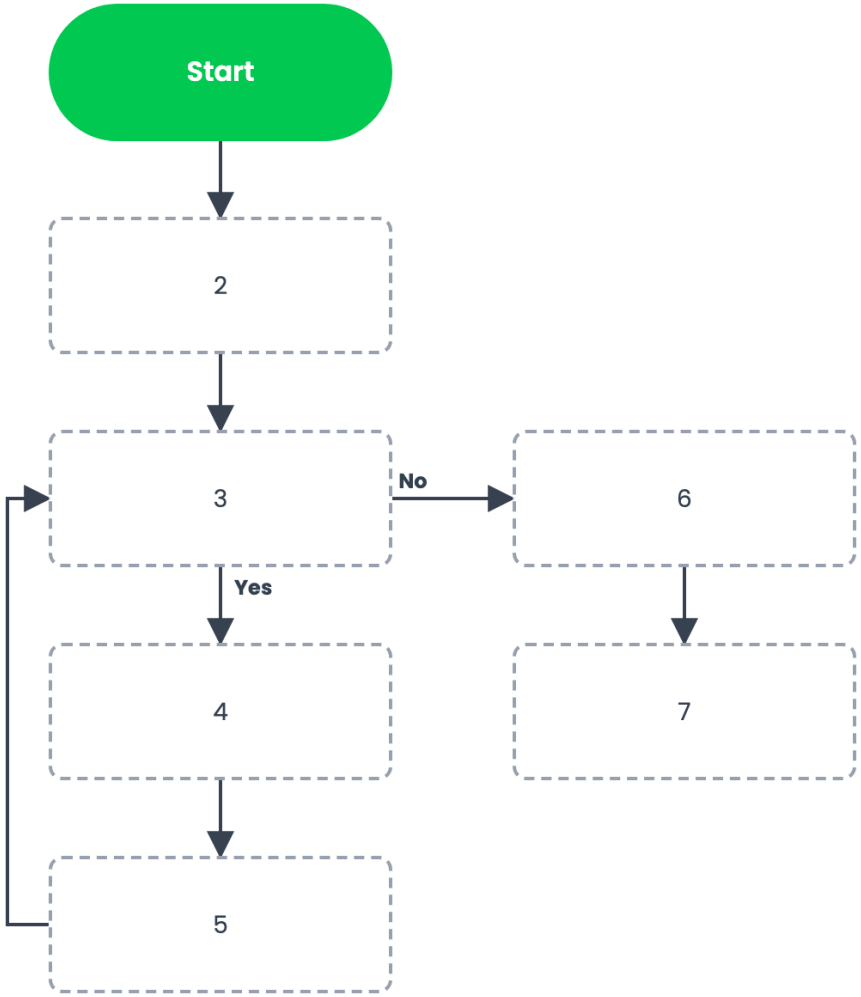


Level 17: Countdown

Medium · Guided

Count down from 10 and finish with a launch message. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



Output "Blast off!"

count > 0?

End

Output count

count = 10

count = count - 1

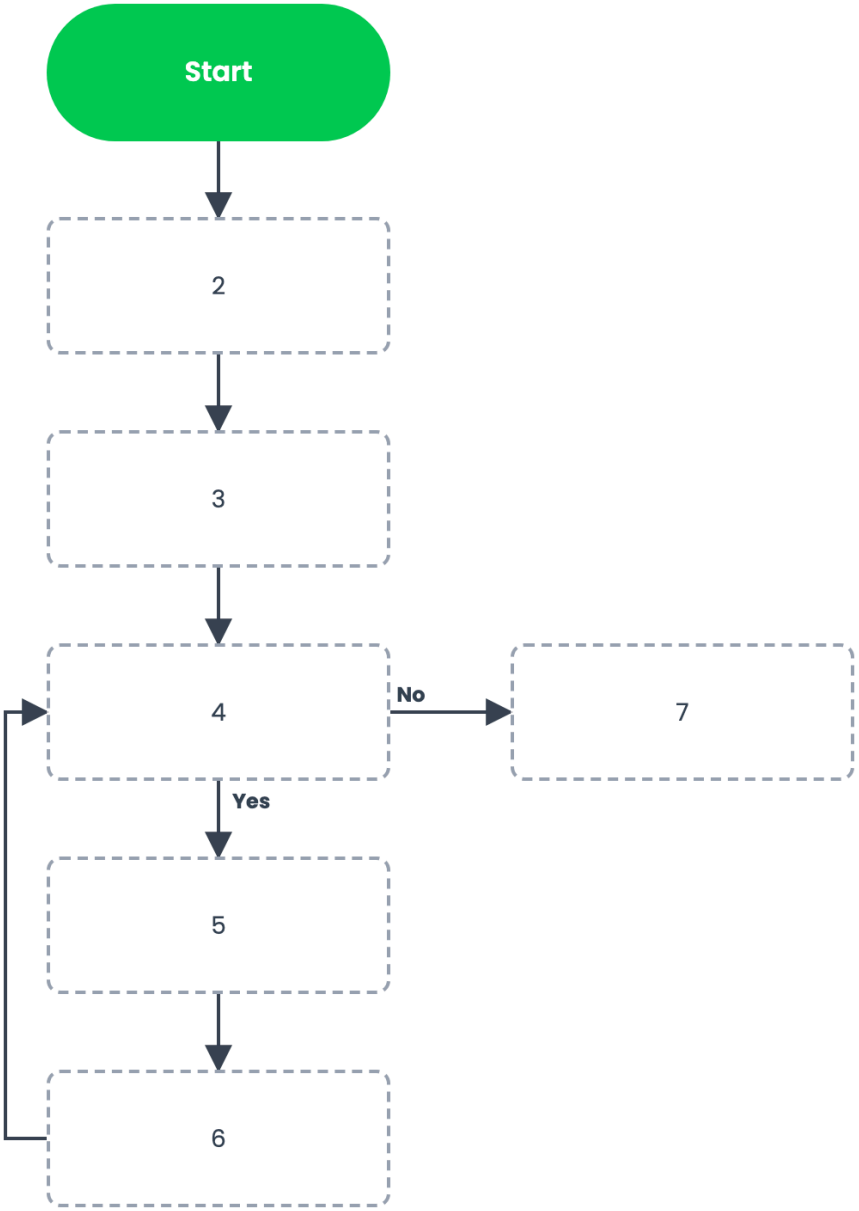
count = count + 1

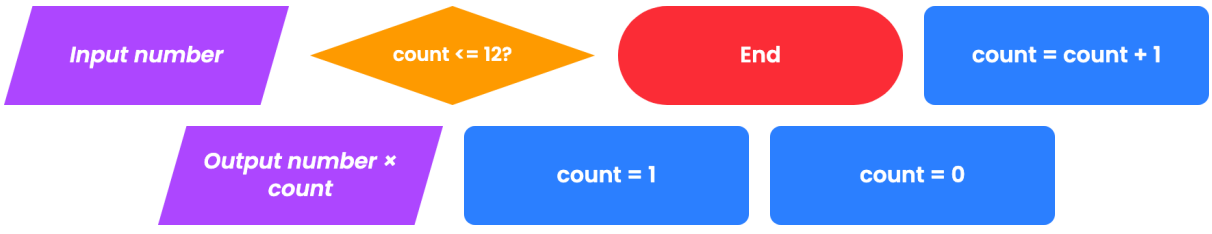
Level 18: Times Table

Medium · Guided

Input a number and output its 12 times table. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



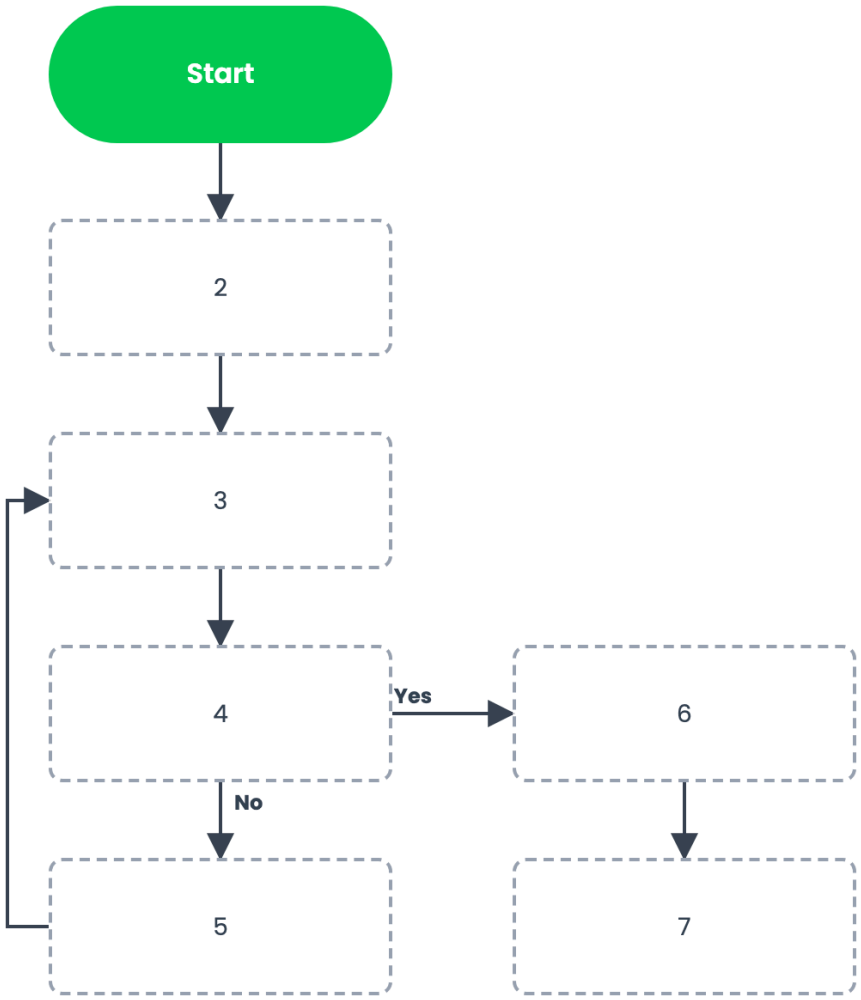


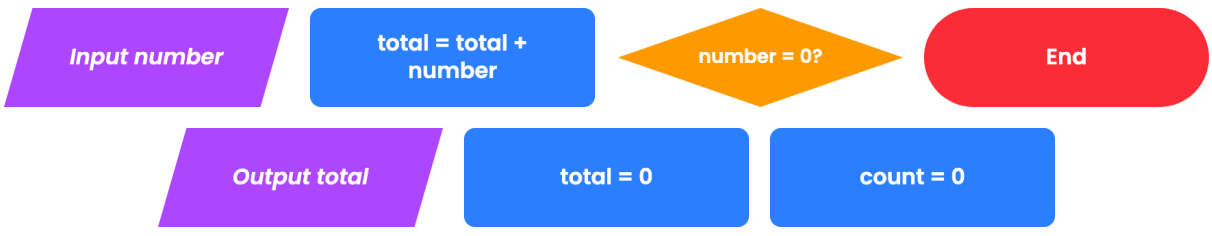
Level 19: Total Until Zero

Medium · Guided

Keep adding numbers to a total until the user enters 0, then output the total. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



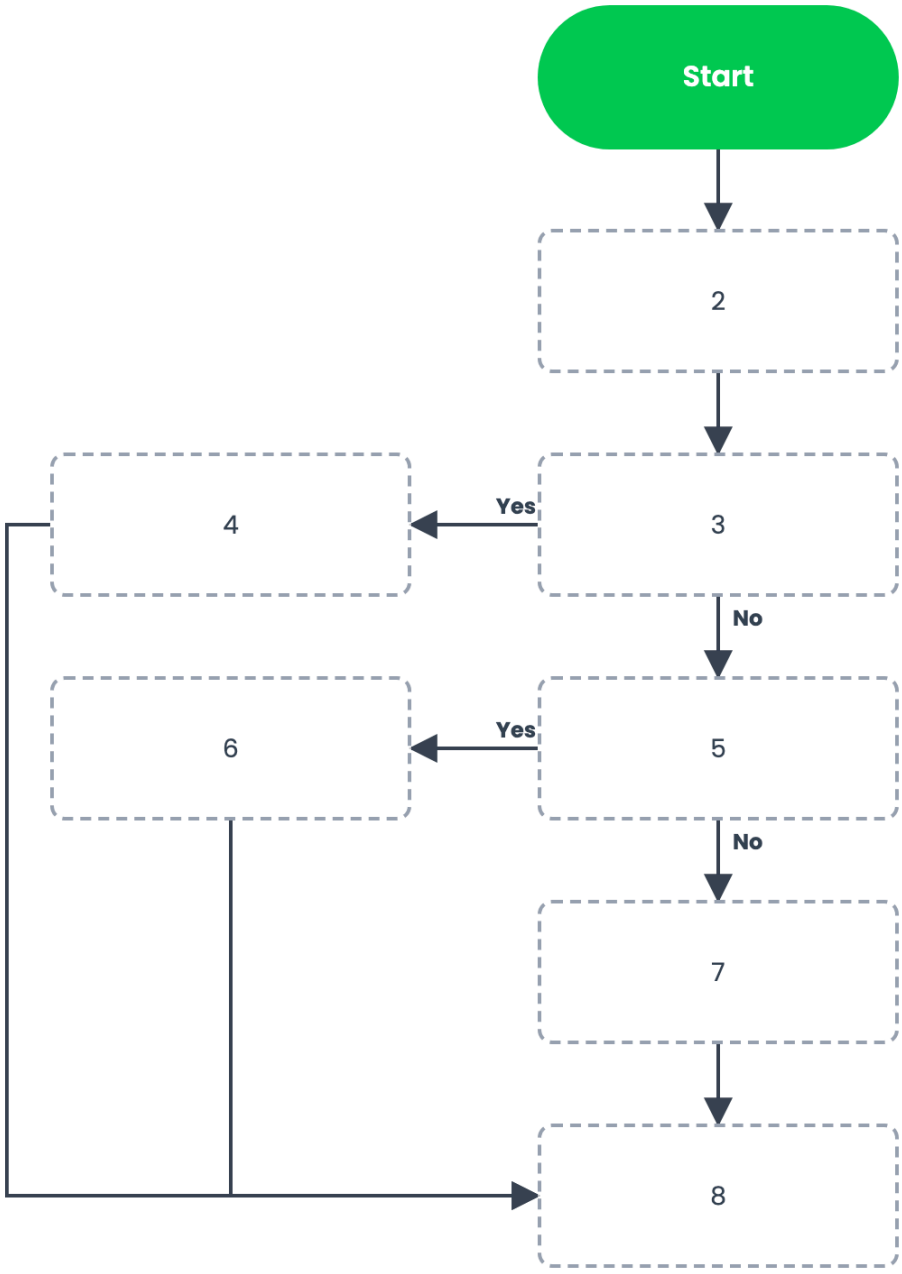


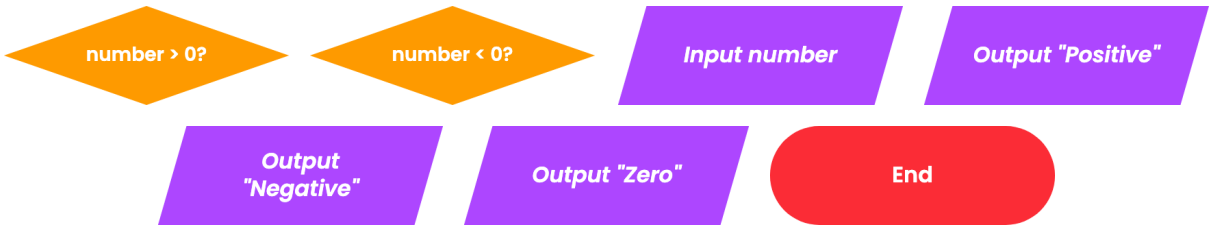
Level 20: Number Sign

Medium · Guided

Output whether a number is positive, negative or zero. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



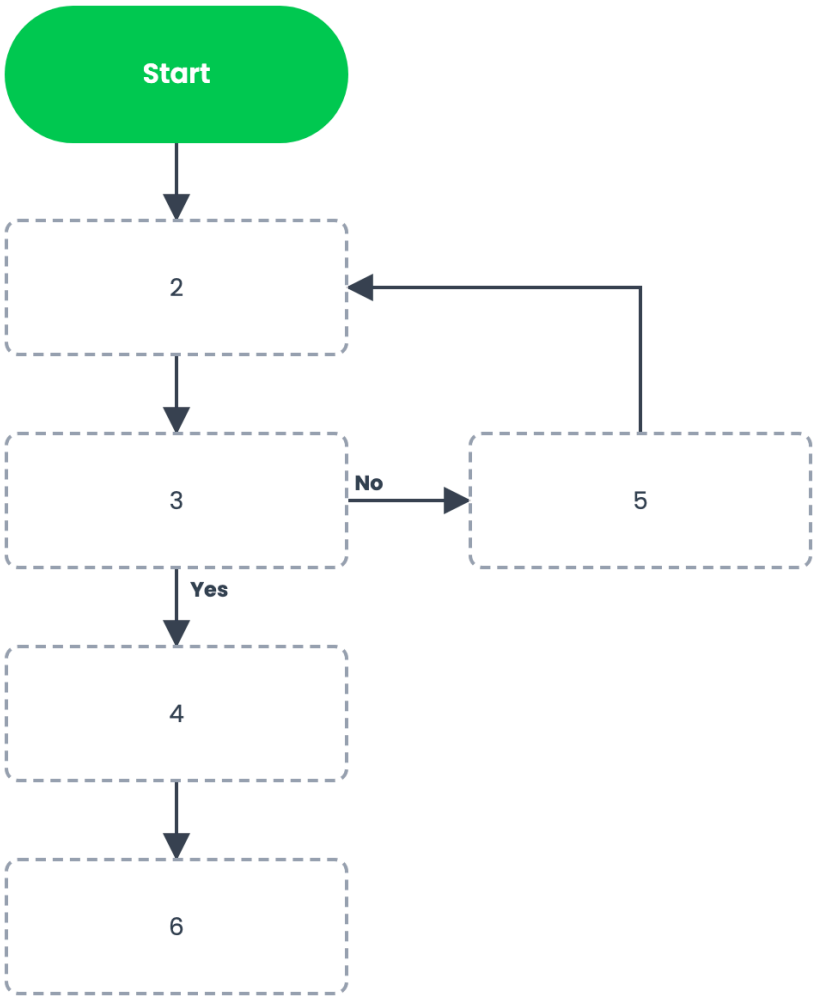


Level 21: Input Validation

Hard · Guided

Ask for a number from 1 to 10. If it is not in range, say so and ask again. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



Input number

number from 1 to 10?

End

Output "Invalid number"

Output "Thank you"

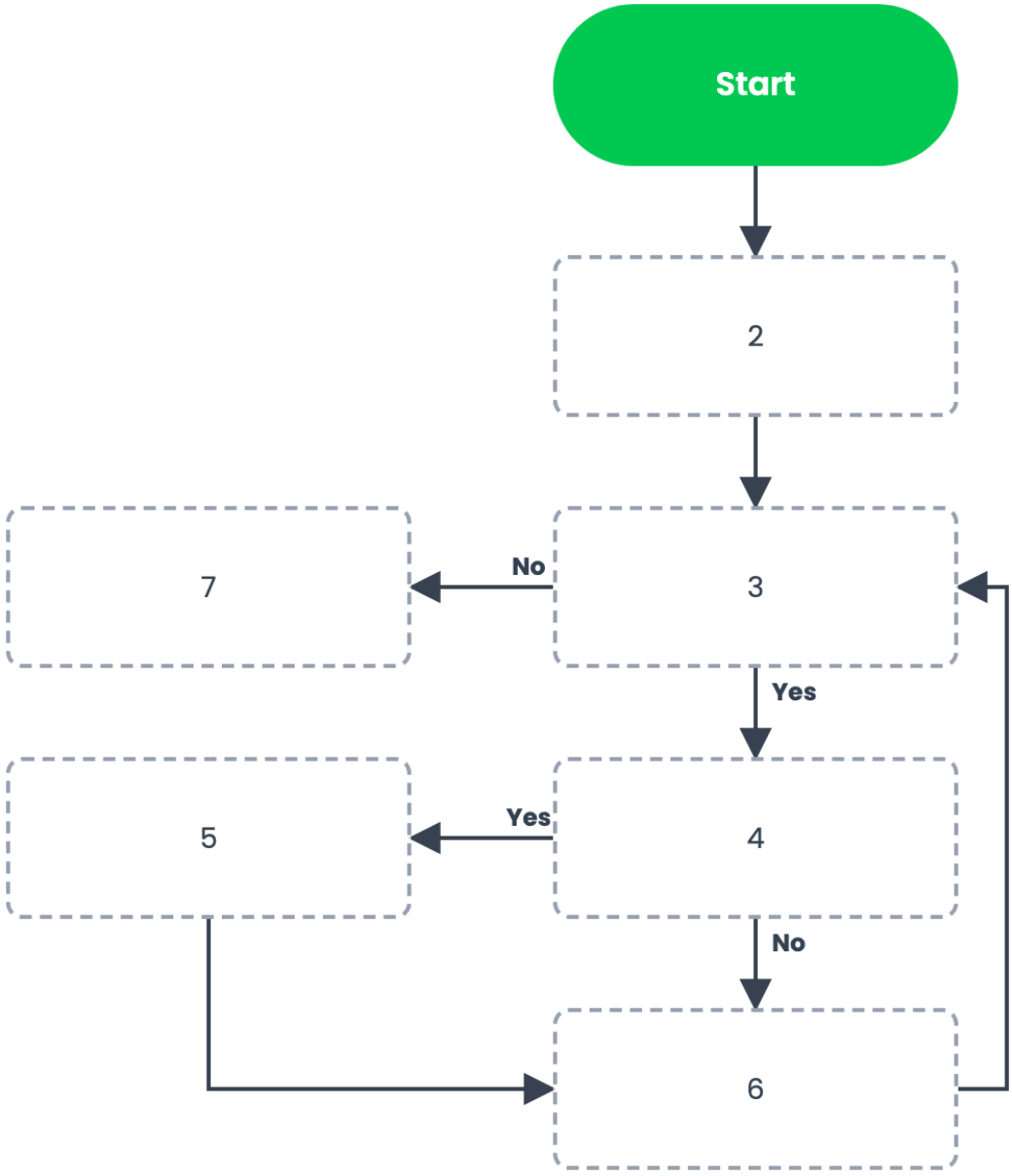
number = 0

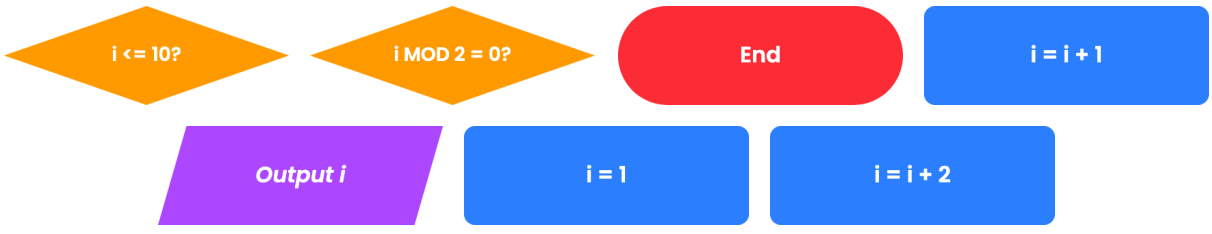
Level 22: Even Numbers

Hard · Guided

Output only the even numbers from 1 to 10. A decision sits inside the loop. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



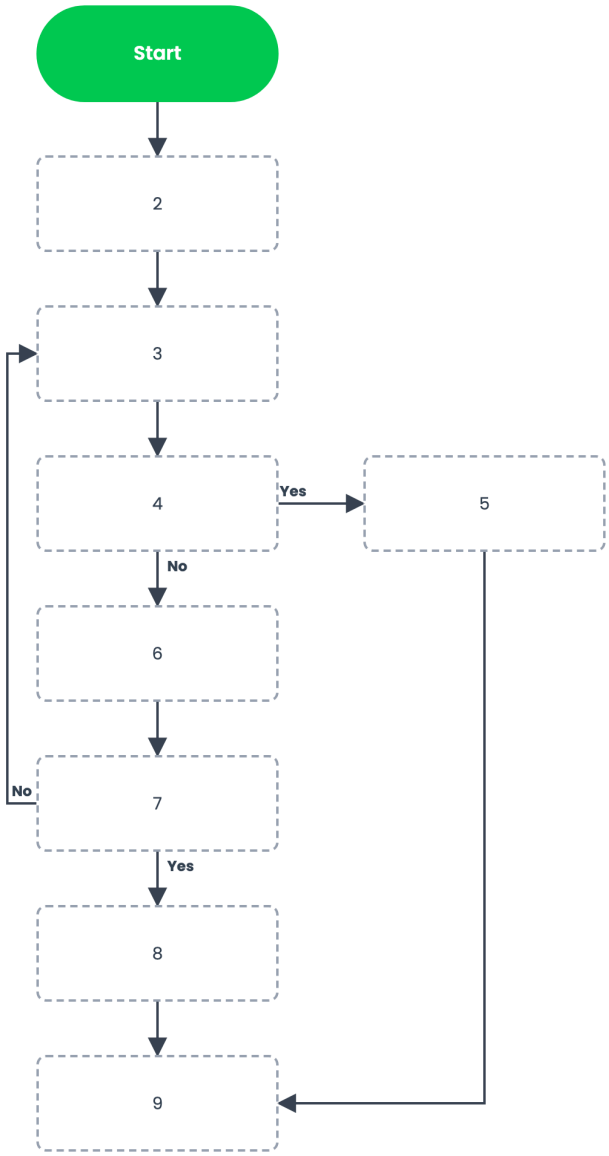


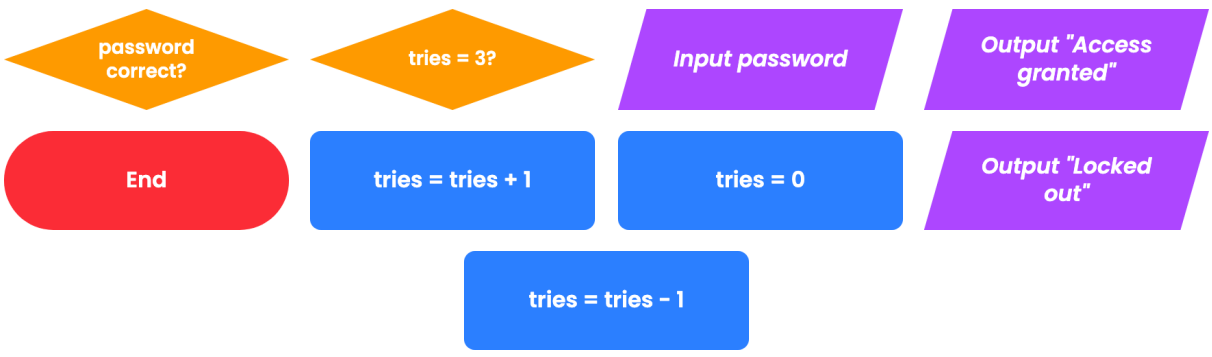
Level 23: Three Attempts

Hard · Guided

The user gets three tries to enter the correct password. After three wrong tries they are locked out. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



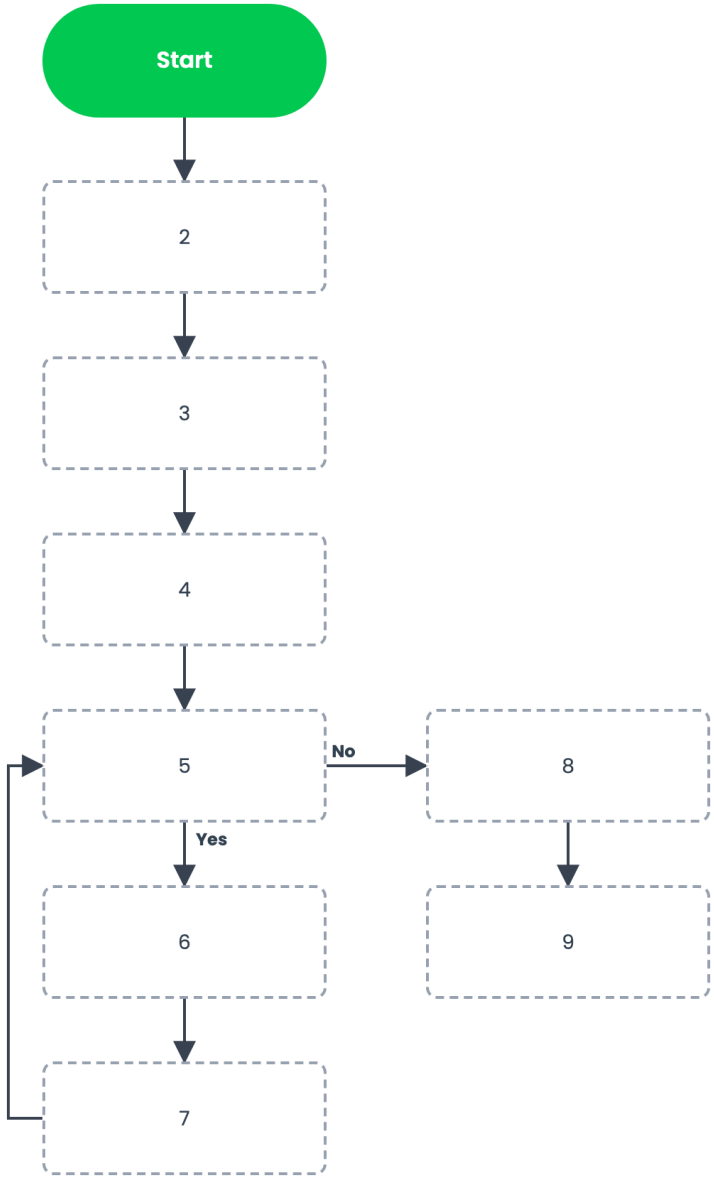


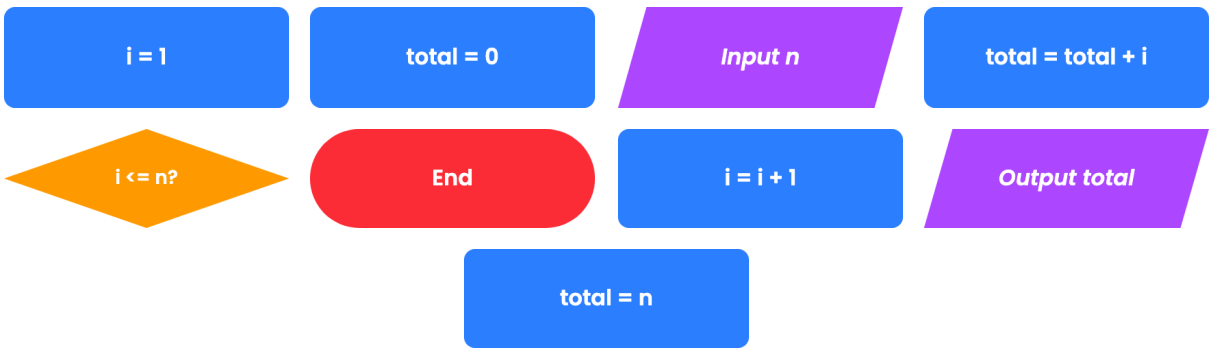
Level 24: Sum to N

Hard · Guided

Input a number n and output the total of 1 + 2 + … + n. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



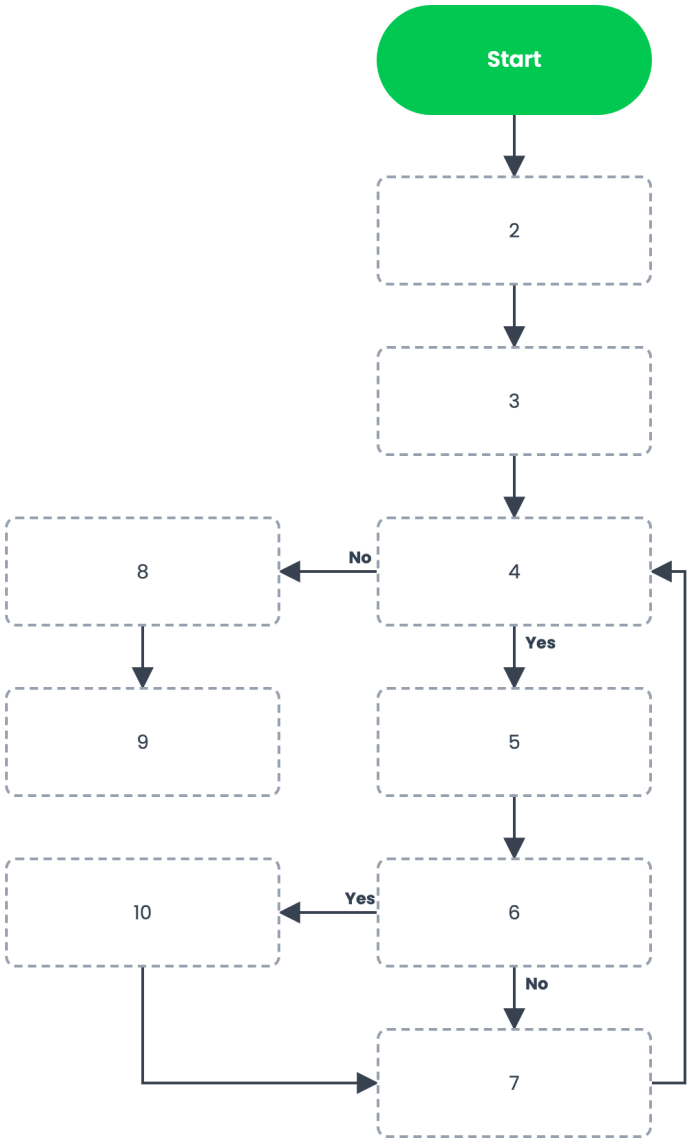


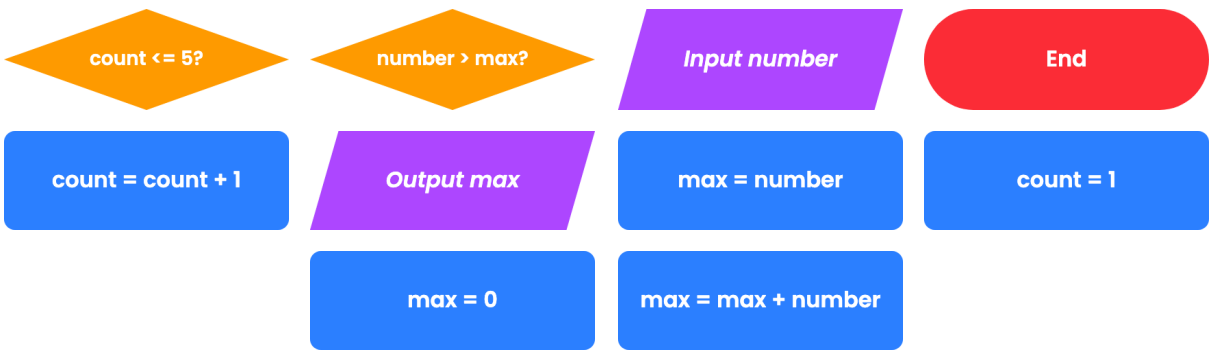
Level 25: Largest of Five

Hard · Guided

Input five numbers and output the largest. Keep track of the biggest so far. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



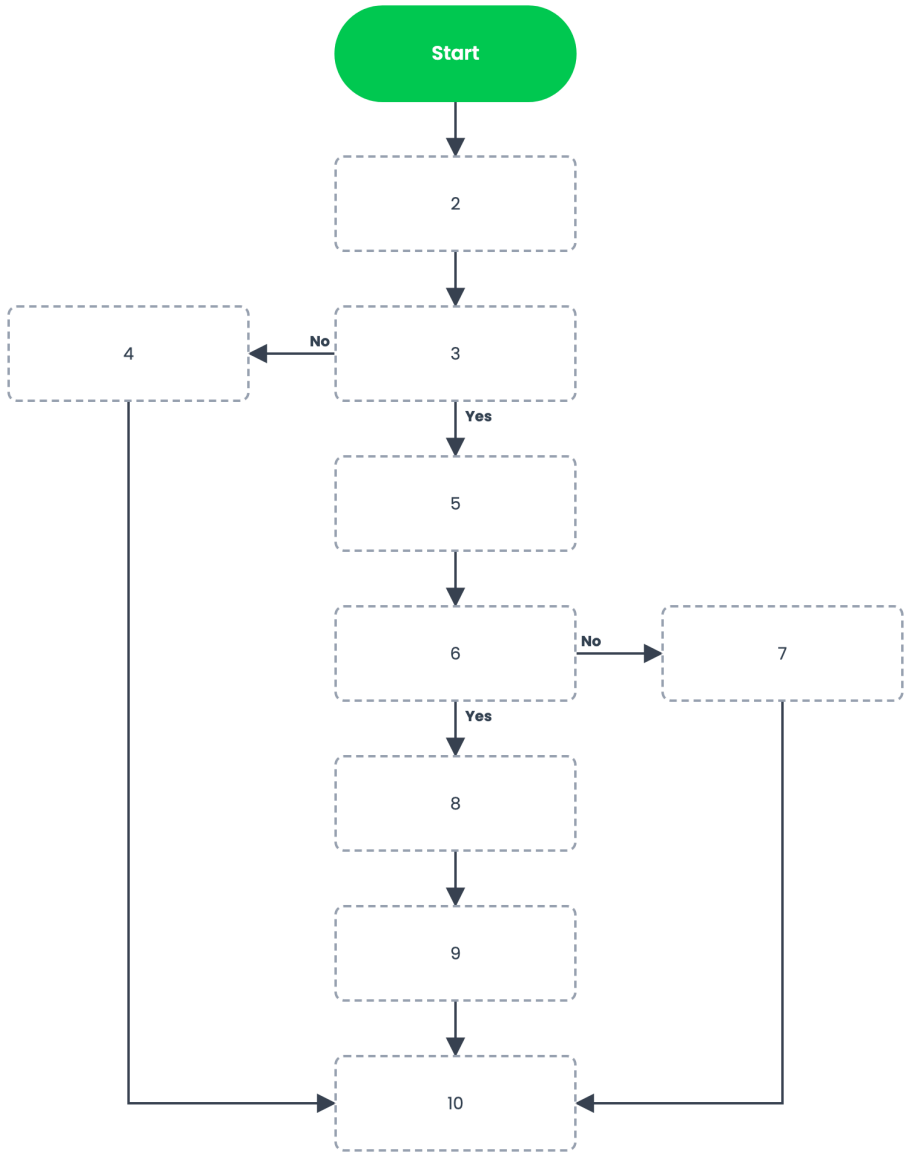


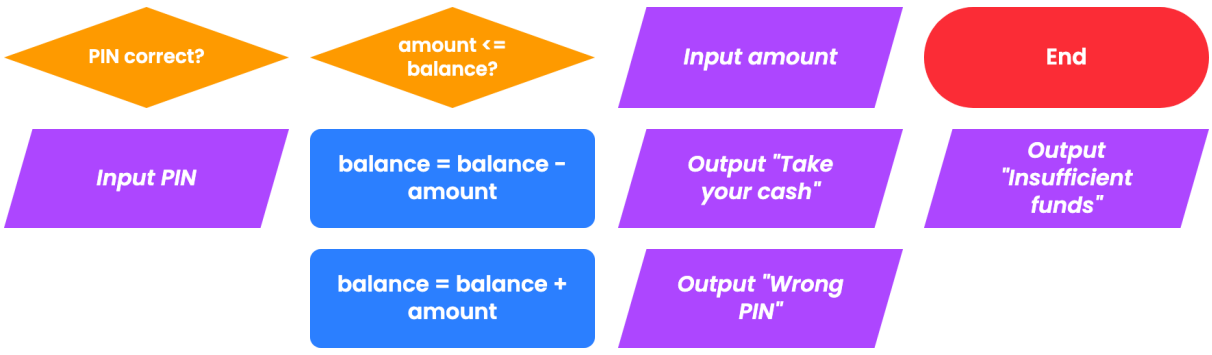
Level 26: Cash Machine

Hard · Guided

Check the PIN, then check there is enough money before paying out. Every path finishes at End. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



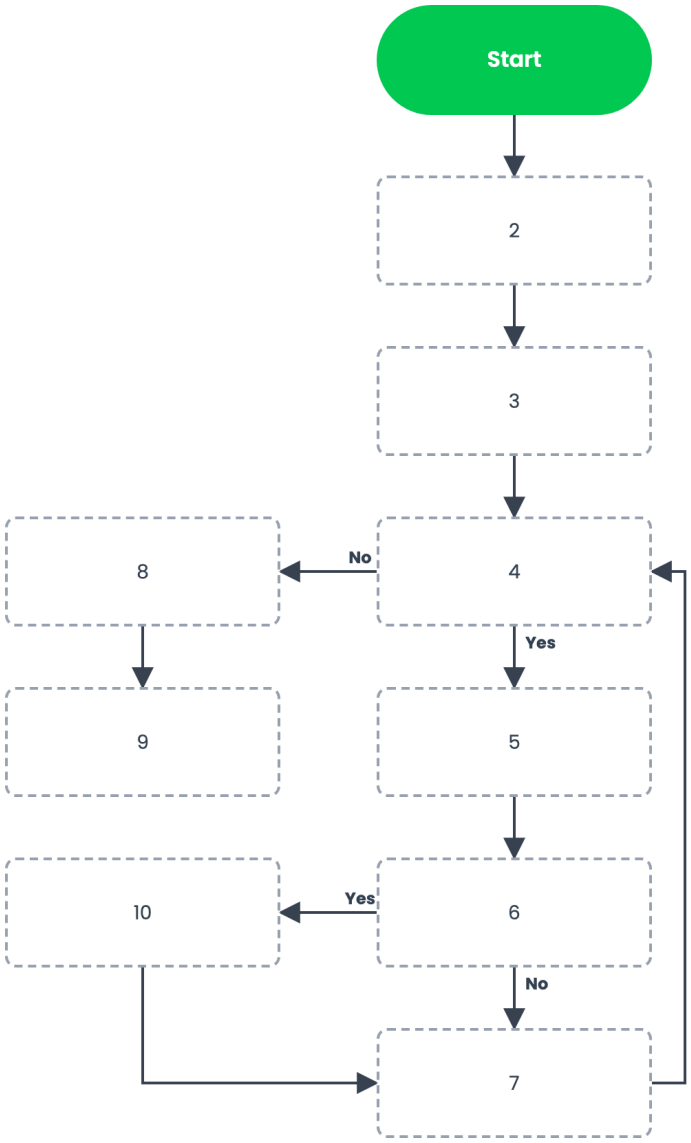


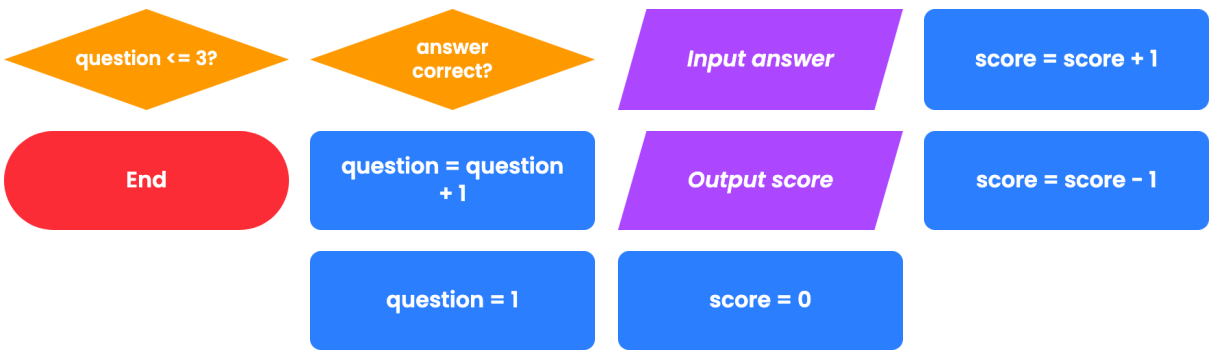
Level 27: Quiz Score

Hard · Guided

Ask three questions. Add one to the score for each correct answer, then output the score. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



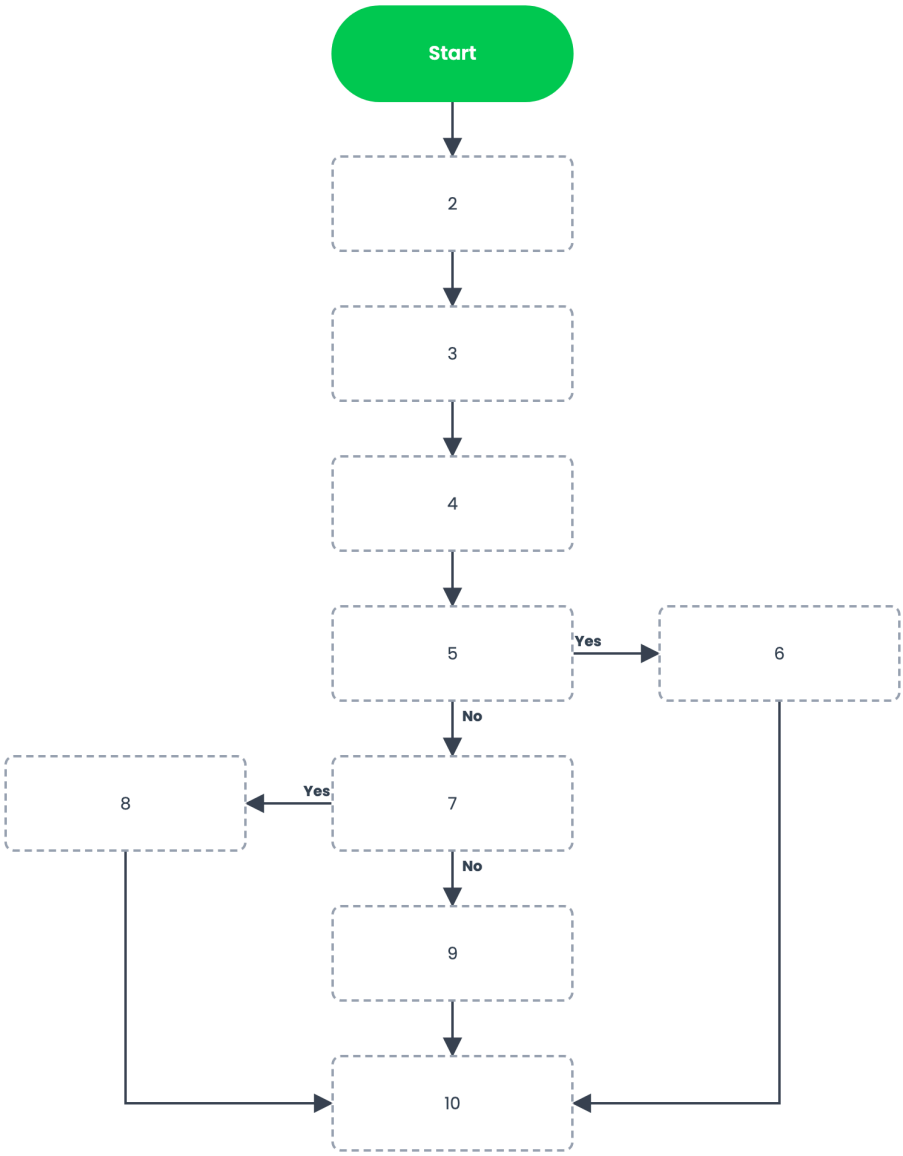


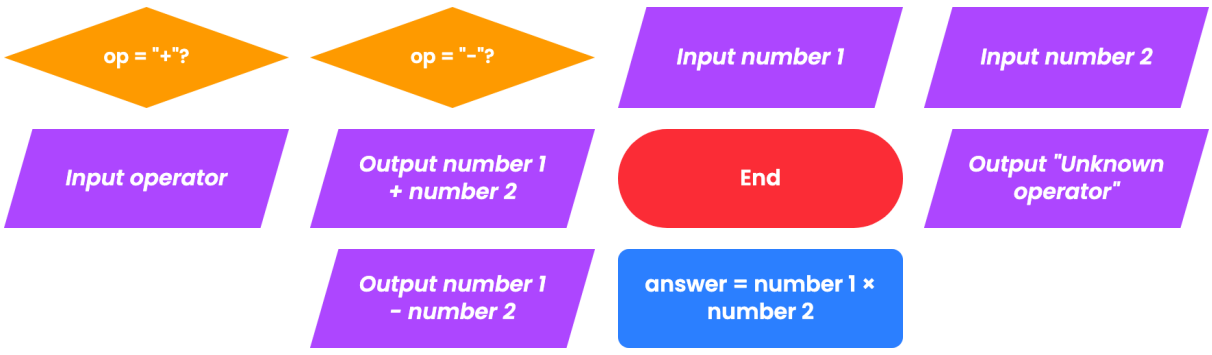
Level 28: Calculator

Hard · Guided

Input two numbers and an operator (+ or -), then output the answer. Anything else is an unknown operator. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.



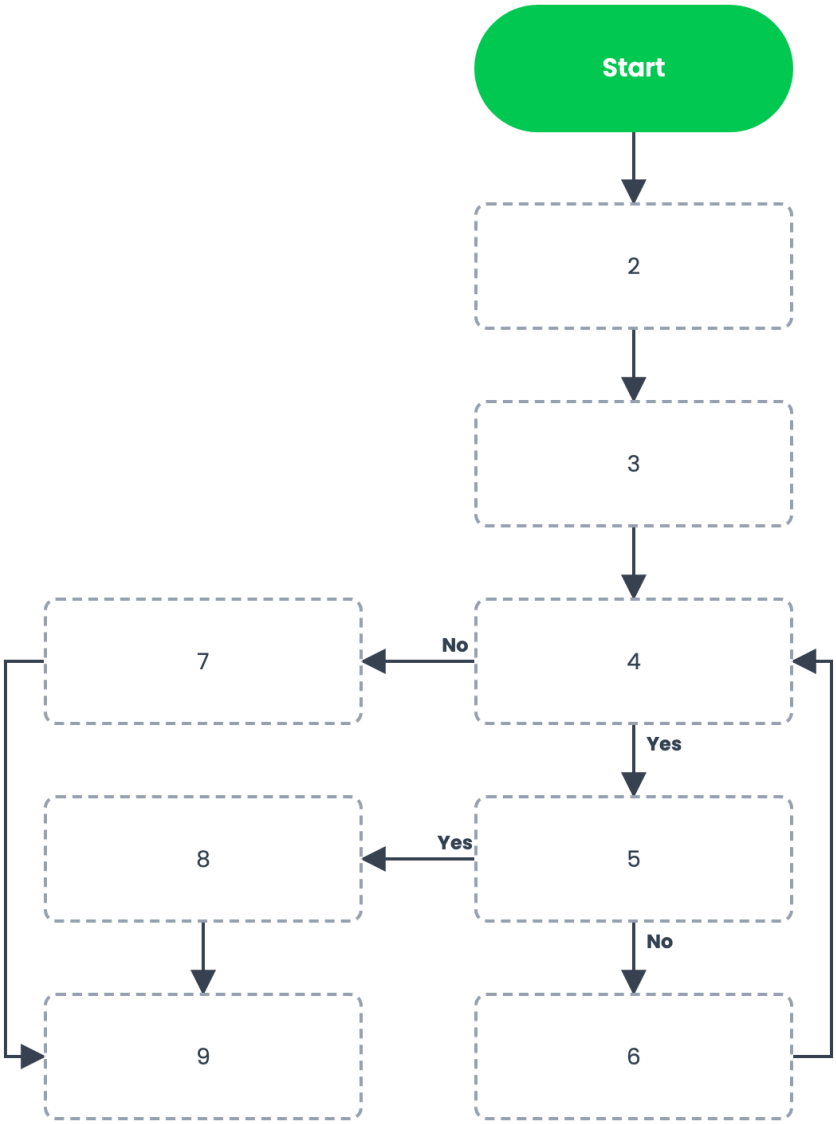


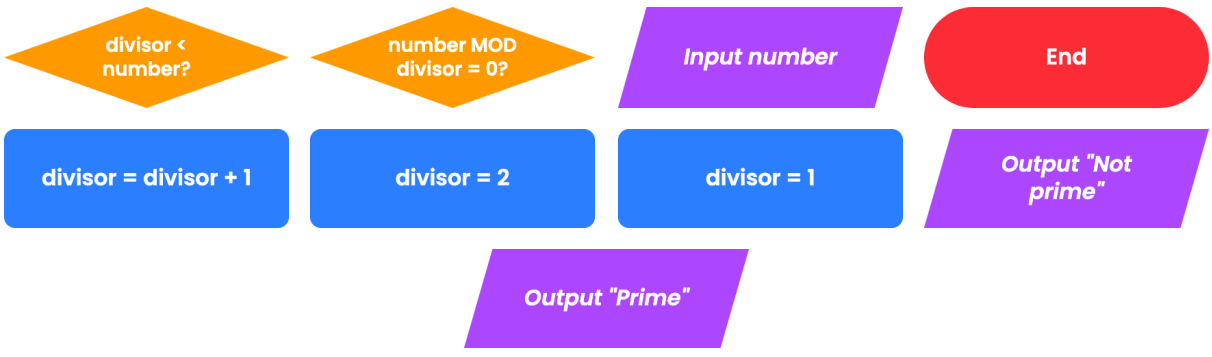
Level 29: Prime Number

Hard · Guided

Try dividing by 2, 3, 4… up to one less than the number. If any divide exactly it is not prime. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.





Level 30: Menu Loop

Hard · Guided

Show a menu: 1 says hello, 2 says goodbye, 3 quits. Any other choice asks again. Start is already placed for you.

The arrows are already drawn. Decide which symbol belongs in each numbered box, then draw it in (or cut out the symbols and stick them in). Some symbols are not needed.

